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RESEARCH ARTICLE

Exhaustivity and contrastivity in Hungarian focus constructions: A visual-world study

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Abstract

Whether the Hungarian pre-verbal focus (PVF) construction encodes exhaustivity semantically or derives it through pragmatic inference has been debated for over four decades. We address this controversy through a visual-world experiment combining sentence–picture verification, self-paced reading, and webcam-based eye tracking with 52 native Hungarian speakers. Participants read sentences in three conditions – explicitly exhaustive (*csak* 'only'), unmodified PVF, and contrastive (*többek között* 'amongst others') – and chose between a single-object (exhaustive) and a multi-object (contrastive) image. Unmodified PVF sentences were interpreted exhaustively in 96.1% of trials, near-identical to the *only* condition, confirming a robust exhaustive default. In the contrastive condition, 62.3% of choices were the multi-object image, showing that a non-exhaustive reading is genuinely available. Crucially, reading times were faster for exhaustive than unmodified sentences at all critical regions, consistent with exhaustivity being pragmatically inferred in the bare PVF. Eye-tracking data showed no significant difference in deliberation time between exhaustive and contrastive image choices in the unmodified condition, indicating both readings remained equally plausible at the point of decision. Selecting the exhaustive image in a contrastive context was significantly costlier in dwell time and fixation count. These findings support a pragmatic account of PVF exhaustivity: the construction is semantically underspecified, yielding a strong but cancellable exhaustive bias while remaining compatible with contrastive interpretations.

Contents

1. Introduction

How languages express contrast and exhaustivity has major implications for how we understand the syntax–semantics interface. This study revisits a theoretical debate on the interpretation of Hungarian focus constructions using eyetracking technology and the visual-world paradigm. Specifically, we pursue two theoretically important inquiries: What are the dominant and available readings of Hungarian pre-verbal focus constructions, and from a cognitive perspective, whether different focus types are processed differently during language comprehension.

(1)

Anna	*meg-evett*	*egy*	*banánt.*
Anna[NOM]	VPX-eat-PST-3SG	a	banana-ACC

'Anna ate a banana.'

(2)

Anna	*egy*	*BANÁNT*	*evett*	*meg.*
Anna[NOM]	a	banana-ACC	eat-PST-3SG	VPX

'Anna ate a BANANA.'

Consider the canonical, neutral sentence in example (1)[1], where the subject *Anna* is in the sentence-initial position, followed by the verb, *evett* 'ate' (together with the verbal prefix [VPX] or verbal modifier [VM] *meg-*)[2] and the object *banán* 'banana' (with the *-t* accusative ending); vs the pre-verbal focus construction in example (2), where the object is fronted to the pre-verbal position, leaving its (normally attached) verbal prefix behind.

[1] Gloss label abbreviations in this and all further examples (except VPX=verbal prefix) are from the Leipzig Glossing Rules (Comrie et al. 2015): ACC=accusative, DEF=definite, NOM=nominative, PL=plural, PST=past tense, REL=relative, SG=singular.

[2] Verbal prefixes in Hungarian can modify the meaning of the verb, or act as aspect markers. *Meg-* in this case has a perfective aspect meaning, indicating not only that the action has ended but also that it is complete, with the apple wholly eaten.

The sentence in example (2) is a typical example for a Hungarian focus construction, and it could also be rendered in English with an *it*-cleft,[3] i.e., 'It is a BANANA[4] that Anna ate.'.

Constituents in canonical sentences can be focalized for various communicative purposes, such as emphasis, question answering, or contradiction. In (2) we are focusing the object, and as such we can observe two changes that mark focus: movement and prosody. Regarding the first, Hungarian has a special, grammaticalized syntactic position for the focused constituent directly in front of the verb – called the pre-verbal focus (PVF) position (Balogh 2007; Wedgwood 2006) or structural focus position (É.Kiss 1998) or simply just focus position – often associated with exhaustive/identificational semantics (Balogh 2013). Besides movement, the syntactically expressed focus is also accompanied by prosodic stress, and it has been suggested that in these focus marking strategies movement is primary, and prosody is secondary (Balogh 2013). Regarding intonation in PVF constructions, the focused constituent typically bears eradicating stress (cf. Gécseg 2020; Kornai & Kálmán 1988), meaning that the verb is destressed and the acoustic prominence of all other elements are leveled. The PVF construction is a common, everyday focus-marking strategy in Hungarian – a topic-prominent, discourse-configurational language where the configuration of constituents expresses information structural roles as opposed to grammatical ones (cf. É. Kiss 1995) – and various aspects of it has garnered significant attention in both theoretical and psycholinguistic research. For a comparative overview on Hungarian focus, see (Bende-Farkas 2006).

1.1. Background

One of the key issues around Hungarian pre-verbal focus is whether it is inherently exhaustive, or if exhaustivity is a pragmatic inference. In other words, does the PVF construction itself entail that the focused element is the only one for which the predicate holds true, or can it be interpreted in a contrastive way, where the focused element stands in opposition to other alternatives without necessarily excluding them? For example, in (2) above, does this sentence implies that Anna ate only a banana *and nothing else* – an exhaustive reading, or could it mean that Anna ate a banana *and not something else* (or even, that she ate a banana but maybe *she also ate something else*) – a contrastive reading? We are interested in the 'default' interpretation of focus constructions such as these, and whether Hungarian speakers tend to understand them exhaustively or contrastively without additional context.

We should therefore try to define what we mean by *exhaustive focus* and *contrastive focus*. First, both types require the pre-verbal structure, but they differ in their semantic/pragmatic implications. In É.Kiss (1998)'s influential paper, both would fall under the term of *identificational focus*, which she defines as a focus that identifies the referent of the focused element from a set of alternatives. It takes scope, binds a variable, and interacts with quantification. É. Kiss treats contrastive focus as a subtype of identificational focus, where she describes it as [+exhaustive], and often [+contrastive]. However, we will argue that identificational focus does not always entail exhaustivity. According to Rooth (1992)'s theory on focus interpretation, exhaustivity is not inherent to focus but arises when focus interacts with specific operators like 'only'. For him, focus is a mechanism that introduces a set of alternatives (*alternative semantics*), and he considers contrast to be the pragmatic use of

[3] While the comparison of the Hungarian focus constructions with English *it*-clefts has received considerable attention in the literature (cf. É. Kiss 1999; Wedgwood et al. 2006) it falls outside the scope of the present discussion.

[4] Linguistically focused elements in the examples will be set in capitals for highlight throughout the paper.

focus, where one alternative is highlighted against others in the set. In Vallduví & Vilkuna (1998), *kontrast* is an operator-like category that introduces a set of alternatives for the focused constituent. If an expression a is contrastive, a membership set $M = \{\ldots, a, \ldots\}$ is generated, serving as a quantificational domain. This allows alternative propositions $P(x)$ to be derived by substituting other members of M for a. Thus, *kontrast* provides the semantic mechanism by which focus evokes alternatives, enabling identificational or exhaustive readings. In Krifka (2008:259)'s work on Information Structure (IS), both exhaustive and contrastive focus is mentioned as subtypes of focus, and while exhaustive is interpreted as the only alternative that makes the proposition true narrowing the set of alternatives down to one (and thus enforcing exclusivity), contrastive focus is defined as a focus used for truly contrastive utterances and presupposes a competing alternative in the common ground, highlighting the contrast. Krifka's distinction aligns with his broader distinction between semantic (truth-conditional impact) and pragmatic (discourse management) uses of focus. Zimmermann (2008) also argued that contrastivity is not a semantic property of focus at all but rather a pragmatic one, and that it is best understood in terms of hearer expectation and discourse predictability, as a discourse-semantic device that signals unexpectedness relative to the hearer's assumptions, rather than a purely semantic operator over alternatives.

In short, exhaustive focus implies that the focused element is the sole entity for which the predicate holds true; i.e. that there are no other things Anna ate. On the other hand, contrastive focus would imply that the focused element stands in opposition to other possible alternatives, drawing attention to its uniqueness, unlikeliness, or contradicting nature given the context, e.g., 'Anna ate a banana [and not an apple].', but it does not necessarily exclude alternatives, i.e., there could be other things that Anna ate, e.g., 'Anna ate a banana [and/but also an apple].' Now consider the following:

(3) *Anna* ***csak*** *egy* *BANÁNT* *evett* *meg.*
Anna[NOM] only a banana-ACC eat-PST-3SG VPX
'Anna ate (up) only a BANANA.'

(4) *Anna* ***többek között*** *egy* *BANÁNT* *evett* *meg.*
Anna[NOM] others amongst a banana-ACC eat-PST-3SG VPX
'Anna ate (up), amongst other things, a BANANA.'

We can, in fact, force an exhaustive reading by adding the exhaustive focus marker *csak* 'only' to the sentence, as in example (3)[5]. One intuitive reading of a simple PVF sentence like that in example (2) would be that 'Anna ate a banana [and nothing else].' In this sense, the sentence seems to be equal to that in (3), where the exhaustive interpretation is explicitly expressed with the operator *csak* 'only'. If we accept É.Kiss (1998)'s view, this seems to be redundant as the structural focus is already exhaustive. The question of differentiating the exhaustivity entailed by the pre-verbal focus structure vs the exhaustivity operator has been studied in detail by Balogh (2007).

Moreover, we can also cancel exhaustivity within the same clause with the help of certain phrases, such as *többek között* 'amongst others', as in example (4)[6], by inserting a phrase with an inherently non-exhaustive, inclusive meaning (see also Balogh 2013;

[5]The scope of *csak* 'only' only extends to the subsequent element, it can only refer to the object and not the following VP.

[6]Similarly, only the object is in the scope of *többek között* 'amongst others', it is not possible to interpret this sentence as if the verb is an alternative to other actions one can do to a banana.

Wedgwood 2006:14). This sentence is not only perfectly grammatical, but inclusive of alternatives and expresses a contrastive sense due the use of focus (there must be some specific discourse function for choosing to focalize a constituent).

In Hungarian linguistic research, the role of semantic exhaustivity in focus constructions has been debated for more than forty years. The central question is whether exhaustivity arises from the syntax–semantics interface or from pragmatic and contextual factors. Earlier accounts within the generative framework assumed that the Hungarian pre-verbal focus structure necessarily entails semantic exhaustiveness, claiming that exhaustivity is an inherent property of the focus structure itself (É.Kiss 1998, 2007; Horvath 2008; Kenesei 2006; Szabolcsi 1981), which was more or less the received view up until the 2000s. More recently, this claim has since been challenged and shown to be untrue by both theoretical and psycholinguistic work – supported by experimental data – and it has been proposed that the exhaustive interpretation emerges from discourse-pragmatics, or conversational implicature (Balogh 2013; Káldi and Babarczy 2016; Onea 2009; Wedgwood et al. 2006). In introducing their argument against the claims of Szabolcsi (1981), É.Kiss (1998, 2007), and Horvath (2008) that 'an immediately pre-verbal focus in Hungarian is semantically exhaustive' or 'is interpreted exhaustively, i.e. as if it were in the scope of an exclusive like "only"', Onea & Beaver (2009:1) suggested in their first example[7] that the implicit reading of a sentence of this type should be exhaustive, i.e. 'Peter kissed Mari and noone else.'. Their example is analogous to our example (2), i.e. 'Anna ate a banana and nothing else.', and their analysis suggests that while such sentences often invite an exhaustive reading, this interpretation is not structurally encoded but contextually inferred.

Thus, looking back at example (2), the underlying implication is not necessarily that Anna ate *nothing* else, but it could be the case that she did not eat *something* else, as illustrated in (5). Therefore, in our view, the interpretation might as well be contrastive. In other words, those who claim that only one of the readings is available to the PVF construction would expect only one of the two readings to survive, contrary to the facts.

(5)	a.	*Anna*	*egy*	*BANÁNT*	*evett*	*meg,*	*(és)*	*semmi*	*mást.*
		Anna[NOM]	a	banana-ACC	eat-PST-3SG	VPX	and	nothing	else-ACC

'Anna ate a BANANA and nothing else.'

[7] *Péter MARIT csókolta meg.*
Peter[NOM] Mari-[ACC] kiss-PST-3SG.DEF
VPX 'Peter kissed MARY.'

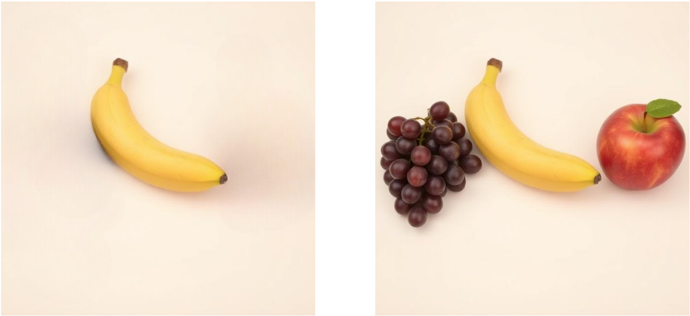

Figure 1. *Example stimuli for a sentence–picture verification task*

b. *Anna egy BANÁNT evett meg, (és) nem egy almát.*
Anna[NOM] a banana-ACC eat-PST-3SG VPX and not an apple-ACC
'Anna ate a BANANA and not an apple.'

To test this, and to address the theoretical debate and in line with recent experimental studies (Gerőcs et al. 2014; Káldi and Babarczy 2018; Káldi et al. 2021; Onea & Beaver 2009), we designed an experiment to investigate how Hungarian speakers interpret pre-verbal focus (PVF), and in particular, whether they distinguish between exhaustive and contrastive focus. Specifically, we aim at answering the following questions: What is the default interpretation of a pre-verbal focus construction? Is it really inherently exhaustive, or could it be contrastive?

2. Experiment

We designed a visual-world experiment with eyetracking in which participants first read a sentence and then chose between two plausible pictures: (A) a single-object image corresponding to an exhaustive interpretation, or (B) a multi-object image corresponding to a contrastive interpretation, see Figure 1 accompanying examples (1–4). Our main question is: How do unmodified PVF sentences (e.g., (2)) pattern relative to the *only*–exhaustive condition (ex. (3)) and the *amongst others*–contrastive condition (ex. (4))? We expect *only* sentences to yield predominantly exhaustive choices and *amongst others* sentences to favor contrastive choices; the crucial question is whether simple PVF sentences will align with one of these conditions or show a more balanced distribution. In addition to the choices, we record reading and reaction times and collect eye movements during the picture verification phase in order to assess whether the conditions have an impact on cognitive processing, and whether different interpretations are associated with different reaction times and/or patterns of attention.

We hypothesize that *only*-exhaustive sentences will be associated with single-object (A) images, with faster responses and more focused attention, while when presented with *amongst others*-contrastive sentences, participants will prefer the multi-object (B) images and may exhibit slower responses and more distributed attention due to the presence of alternatives and the ambiguity of the underlying meaning. This would suggest that the PVF construction in itself is not exhaustive.

2.1. Participants

We recruited 55 native speakers of Hungarian via the Prolific platform (www.prolific.com), where participants were pre-screened for Hungarian as their first language, and received a monetary compensation for their time (around £7.50/h). After rejecting 3 participants (due to overly long submission time, failing attention checks, or missing ET data), the final sample consisted of 52 individuals (32 female; mean age = 35). On average it took people just under 23 minutes to complete the experiment, although with a high individual variation (unusually long submission times were rejected).

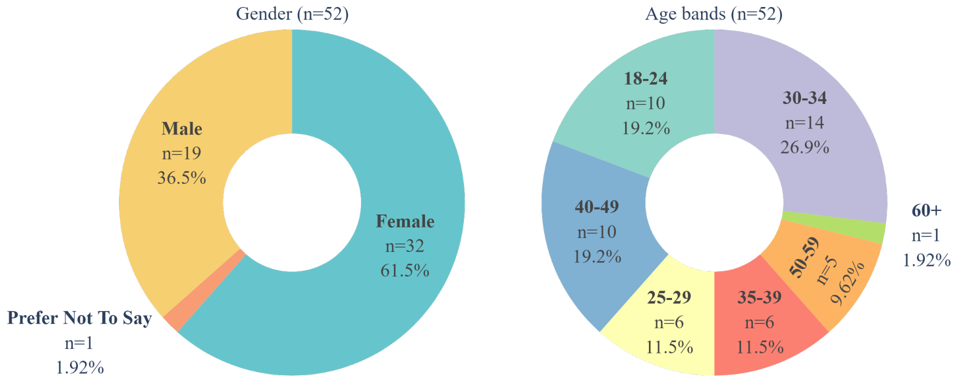

Figure 2. *Demographic information of the participants in the experiment: gender (left) and age distributions (right)*

2.2. *Materials*

Our stimuli consisted of 54 experimental sentences (18 items × 3 conditions) with 54 additional filler sentences in a randomized order (108 in total), but making sure that the same items never followed each other. Regarding lexical variation, we used verbs both with and without verbal prefixes (50-50%), and we included instances of both definite and indefinite conjugation (44.4-55.6%); also, both countable and uncountable nouns were used for object focus. We added some simple contextual fillers to the beginning of each sentence, such as *ami X-t illeti* 'regarding X' and time or place adverbials to balance sentence length and make the targeted focus constructions appear not in the beginning of trials. When presenting the sentences on the screen, we divided them into regions, and articles were kept together with nouns, so all three conditions had the same number of regions (7), and the focused element was always in region 6 (r6). The stimuli were created by changing a set of 18 simple transitive sentences by modifying the base sentence with the addition of the focus marker *csak* 'only' for the exhaustive condition, and the phrase *többek között* 'amongst others' for the contrastive condition.

The 18 items were accompanied by 18 image pairs, these were created using DeepAI's image generator (https://deepai.org/) with prompts requesting simple 3D objects & scenarios in front of a light pastel background, and then modified manually to get matching pairs. The full stimuli list is available online[8] and in the Appendix (see the sentence list in Table 6.1 and the image pairs in Fig. 12).

To avoid overly long experimental sessions, we split the experiment into three equal parts and each participant could complete one, two, or all three parts as separate events; meaning that we had three participant groups that could overlap, and each experiment consisted of 40 trials (36 experimental + 4 attention checks). This resulted in 72 valid submissions (25 for group one, 23 for group two, and 24 for group three), thus every stimuli was interacted with at least 23 times. The three parts were split in a way that for each group each item only appeared in one condition, and each session contained four attention check questions intermixed and evenly distributed across the trials.

[8] OSF link: https://osf.io/j9abg/overview?view_only=484a4f21a5cb4e51b4b826f970e096a

The three conditions were: a) exhaustive focus with *csak* 'only' (e.g., (6a)), b) unmodified, simple pre-verbal focus (PVF) (e.g., (6b)), and c) contrastive focus with *többek között* 'amongst others' (e.g., (6c)). Each item was paired with two images: A) one depicting a single-object interpretation – the exhaustive reading – and B) one depicting a multiple-object interpretation – the contrastive reading. See Figure 3 for an example of the image pairs accompanying the sentences in the experiment.

Figure 3. *Image pair for the sentence in examples (6a–6c). The left image (A) corresponds to the exhaustive reading, while the right image (B) corresponds to the contrastive reading.*

(6) a. **exhaustive focus**

Ami az innivalót illeti, Eszter csak kávét ivott.
which[REL] the drink-ACC regard-3SG Eszter only coffee-ACC drink-PST-3SG
'Regarding drinks, Eszter only drank coffee.'

b. **unmodified pre-verbal focus**

Ami az innivalót illeti, Eszter ma kávét ivott.
which[REL] the drink-ACC regard-3SG Eszter today coffee-ACC drink-PST-3SG
'Regarding drinks, Eszter drank coffee today.'

c. **contrastive focus**

Ma reggel Eszter többek között kávét ivott.
today morning Eszter others amongst coffee-ACC drink-PST-3SG
'This morning, Eszter drank, amongst other things, coffee.'

2.3. Design

We built a sentence–picture verification (SPV) task, in which participants read a sentence and were asked to indicate the best fitting meaning-representation between two visual stimuli. First, participants saw a preview of the image pair to get familiar with the objects/context, on the left and right half of the screen (20% of the screen's width and 40% of the screen's height) on a canvas 50% larger than the images to mitigate the inaccuracies of the eye tracker. After pressing the spacebar they were presented with the sentence word by word (region by region) in the middle of the screen in a self-paced reading (SPR) manner, in which participants had to use the spacebar to advance throught the sentences. We recorded reading times throughout. After reading through the sentence, the image pair re-appeared and participants had to select the image that best matched the meaning of the sentence they just read. The choice had to be made with a mouse-click anywhere on the corresponding canvas (left or right), and reaction times (RT) and eye movements were recorded during this picture verification phase. The image types were counterbalanced throughout the trials. The segments were separated by 1000 ms pauses, and after the choice, the next trial started after a brief (1000 ms) pause. The experiment was designed to be completed in around 25-30 minutes, and participants had to complete it in one setting. See Figure 4 for a visual representation of the design of the experiment.

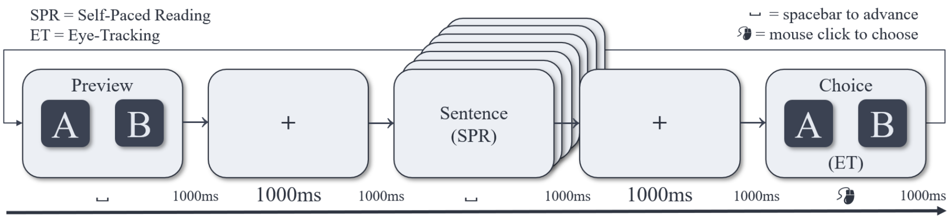

Figure 4. *Design*

2.4. Procedure

The experiment was created, hosted on, and presented to participants on PCIbex Farm (Schwarz & Zehr 2021),[9] in Hungarian. Participants completed the experiment on their own devices (PC only, with a webcam and physical keyboard & mouse) and on their own time, but with Prolific helping us control for too fast or too slow submissions resulting in rejection/timeout.

First, potential participants were instructed to be in a well-lit environment, sit at an appropriate distance from their screen and webcam, and minimize head movements; emphasizing the need for a bright location where there faces are clearly visible to the camera. They were also informed that they can only start the experiment from a Chrome browser. Before starting the experiment, participants then had to calibrate the eye tracker, which involved looking at specific points on the screen (e.g., corners, center) while the system records their eye positions to create a mapping between gaze coordinates and screen coordinates. Eye-tracking data were collected using the built-in WebGazer.js library of PCIbex, which uses the participant's webcam to track eye movements, and we used the default calibration procedure provided, a 9-point calibration grid. Participants had to reach and maintain at least 60% accuracy in the calibration phase to proceed with the experiment, and had three attempts to achieve this accuracy, ensuring that the eye-tracking data collected during the picture verification phase would be reliable. The tracker also checked

[9]Code and resources are also available online for inspection: https://osf.io/j9abg/overview?view_only=484a4f21a5cb4e51b4b826f970e096a

their gaze repeatedly before every trial, and if the accuracy dropped below 60%, participants were prompted to recalibrate before proceeding to the next trial. People who could not pass the calibration for various reasons (e.g., computer setup, bad lighing/environment, glare on glasses, or any other technical issue) were not able to start the experiment. After calibration, participants were intstructed on the procedure and how to complete the task, followed by a practice session consisting of three trials, and finally they could began the live experiment of 40 trials (including 4 attention checks).

We have to mention that a relatively large amount of people (around a third of all who tried) could not pass calibration, and thus were not able to start the experiment. This is a persistent issue with webcam-based online eye tracking studies, as it can be affected by various factors such as lighting conditions, webcam quality, and even individual differences in facial features. Depending on these factors, participants' experiences were vastly different, some people breezed through trials and had a smooth experience finishing under 20 minutes, others struggled and had to recalibrate from time to time, spending over up to 40 minutes in total. We urged participants to reach out if they had issues, but sadly we couldn't assist everyone remotely. A small number of participants started but couldn't finish the experiment due to technical reasons, they were also compensated for their time, but their data was not included in the final analysis.

2.5. Analysis

2.5.1. Eye-tracking data processing

Eye-tracking data recorded on PCIbex comes in a relatively straightforward format not of gaze coordinates but in classified samples, where each sample is already defined as a gaze at a canvas or outside the canvas, in our case: left canvas, right canvas, or neither. Samples are recorded between every 30-70 milliseconds, depending on the users' computer & browser setup, internet speed, and presumably the general load that the PCIbex server was under at the time.

The following steps were taken to clean and classify the eye-tracking data in order to obtain a certain set of metrics that could be used for analysis: First, raw eye tracking data files were loaded per participant, and **samples** were classified into AOIs at each timestamp as left (left canvas=1, right canvas=0), right (right canvas=1, left canvas=0), or neutral otherwise. Then, same-side consecutive samples were collapsed into contiguous gaze **episodes** within participant and trial. Episode duration was computed from timestamp-derived sample durations where episodes shorter than a threshold of 60 ms were removed, and neutral-gap bridging was enabled (with a max neutral bridge of 50 ms). In other words, very short gaze episodes and rapid saccades were removed cleaning noise & jitter, as they are unlikely to reflect meaningful cognitive efforts and more likely to be artifacts of the eye-tracking system. Then, neutral episodes were dropped and excluded from further analysis. Episodes thus intend to represent a moment of sustained gaze within an AOI, ignoring brief out-of-canvas samples that result from technical inaccuracies and are not deliberate (or uncounscious) eye movements. For quality control, episode-duration distributions were inspected via summary statistics, and episodes longer than the mean +2 SD ($\mu \pm 2\sigma$) were winsorized by replacing their duration with the global mean (i.e., with rows retained). This procedure preserved the full number of episodes while limiting the influence of extreme outliers; 4.89% of all episodes were substituted. See Figure 5 for the distribution of episode

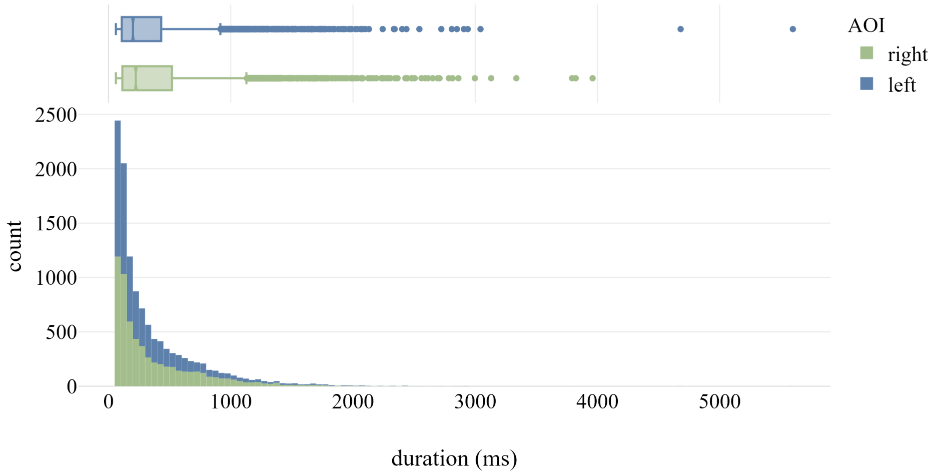

Figure 5. *Distribution of episode durations after cleaning and before winsorization*

durations. This cleaning process was devised after established eye-tracking data classification protocols, such as the Identification by Dispersion-Threshold (I-DT) algorithm, which itself is ideal for low- to mid-frequency eye trackers and static stimuli.

Finally, trial-level eye-tracking (ET) features were computed, including left/right/total dwell time, dwell proportions, dominant side, fixation counts, revisits, transitions, and first-fixation side/duration. To briefly define the most important metrics that were used in the end: **dwell** is a time value, the sum of the duration of all gaze episodes within a trial (either left, right, or total), representing the total amount of time a participant has spent looking at the AOI(s) during the choice; **fixation** on the other hand is a numeric value, representing how many times a participant looked at the AOIs during a choice. Missing trial-level metrics were mean-imputed in 5.02% of the data points, which were mostly due to missing episode-level data (e.g., no episodes detected for a trial, or no episodes remained after cleaning).

We acknowledge that several processing decisions may limit inference. Gaze data was essentially collapsed to binary AOIs (left/right), while off-target samples were treated as neutral and excluded. Episodes shorter than 50 ms were removed, potentially excluding very brief but real events. Long-duration episodes were winsorized by replacing values above a global mean+2SD threshold with the global mean, preserving sample size but reducing variance and potentially biasing duration-based measures. Also, missing merged ET trial metrics were mean-imputed for selected variables, which further shrinks the effective sample size. We therefore interpret absolute magnitudes cautiously and focus on relative contrasts between conditions. The eye-tracking data cleaning and classification pipeline was implemented in Python, and can be inspected online.

2.5.2. Statistical analysis

We analyzed two eye-tracking dependent measures: *total dwell time* (`total_dwell`) and *total fixation count* (`total_fixation`). For both measures, analyses were conducted in R using linear mixed-effects models (LMM) (packages `lme4`, `lmerTest`, `car`, and `emmeans`). The fixed-effect predictors were focus Condition (three levels: exhaustive, unmodified, contrastive) and image Choice (two levels: single-object, multi-object). To stabilize variance and accommodate zero values, we applied a log transform using $\log(1 + x)$ to the raw dependent measure prior to modeling. For each dependent measure, we fit the following model with Participants and Items as random intercepts. The formula is:

```
y' ~ Condition * Choice + (1 | Participant) + (1 | Item)
```

where $y' = \log(1 + y)$ is the transformed dependent variable. The models were fit using maximum likelihood estimation (i.e., `REML = FALSE`) to enable likelihood-ratio model comparisons. To follow up on significant effects in the model, we computed estimated marginal means and conducted Tukey-adjusted pairwise comparisons using `emmeans`. Specifically, we tested (i) pairwise differences between conditions within each type (Condition | Choice), (ii) the pairwise differences between types within each condition (Choice | Condition), and (iii) the full interplay of both (Condition $*$ Choice).

3. Results

3.1. Behavioral results

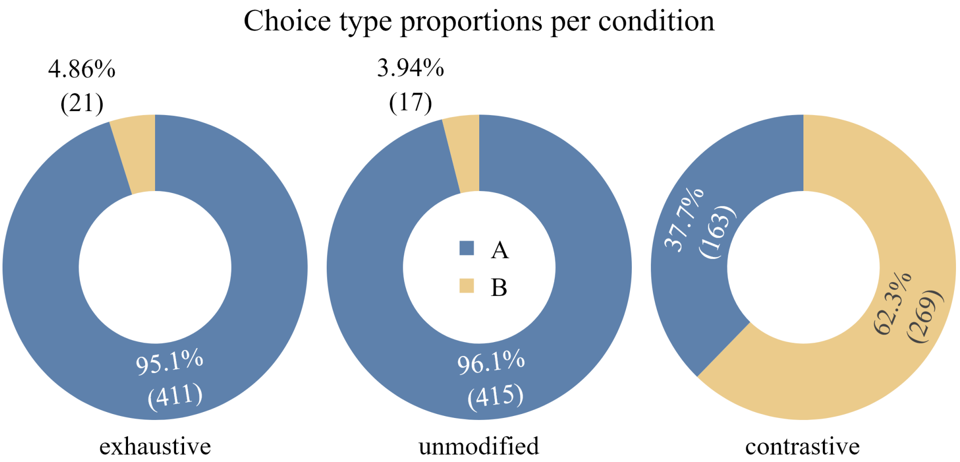

Figure 6. *Distribution of chosen visual stimuli (Image type A vs B) per condition*

The behavioural results showed that participants overwhelmingly interpreted the unmodified pre-verbal focus (PVF) sentences like exhaustive *only*-sentences, choosing single-object type A images. However, in case of contrastive *amongst others*-sentences,

Table 1. *Type of chosen image per condition.*

Condition	Image A: % (N)	Image B: % (N)
Exhaustive	95.1 (411)	4.86 (21)
Unmodified	96.1 (415)	3.94 (17)
Contrastive	37.7 (163)	62.3 (269)

people preferred the multi-object type B images, exhibiting a tendency toward non-exhaustive readings, but with almost 37.7% of the choices – somewhat unexpectedly – still being cast on the exhaustive interpretation type A images (see Table 1, Figure 6). This suggests that even with the presence of a phrase that licenses an inclusive, contrastive meaning, the exhaustivity effect is still quite salient. This pattern provides clear evidence that the PVF construction is not obligatorily interpreted as exclusively exhaustive; a non-exhaustive interpretation remains available. Our results further show that a contrastive reading can be elicited readily, and that the presence of an inclusive, contrastive licensing phrase does not fully eliminate the construction's exhaustive bias.

3.2. *Reaction Times*

We analyzed the reading times during self-paced reading and reaction times (RT) during the picture verification phases, see the plots in Figure 7 for an overview. Reading times and decision reaction times were winsorized by replacing values above the mean +2 SD with the mean to mitigate the impact of extreme outliers. In the following analyses, values were log-transformed using $\log(1 + x)$ to stabilize variance.

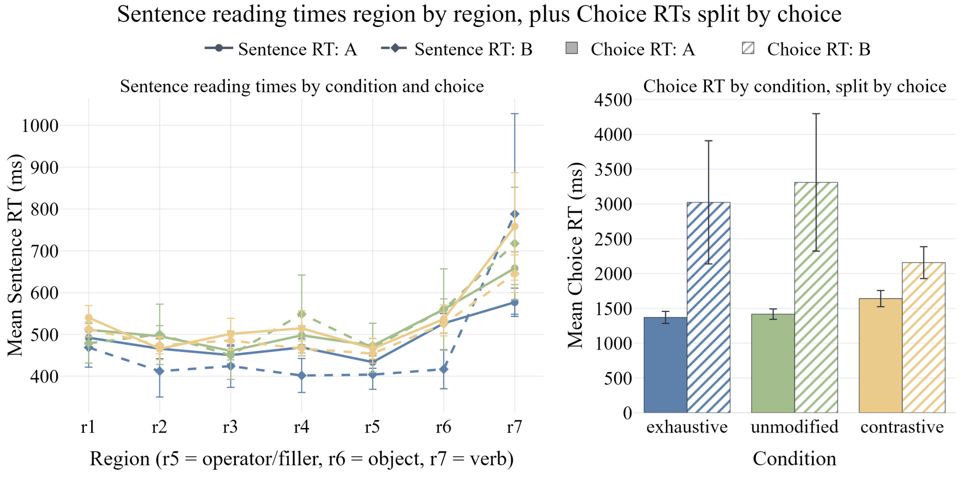

Figure 7. *Mean reading times of sentences by regions and reaction times of choices.*

3.2.1. Sentence Reading Times

Table 2. *Type III Wald chi-square tests for whole-sentence reading times.*

Effect	χ^2	df	p
Condition	18.66	2	< .001
Choice	0.06	1	.806
Condition × Choice	2.10	2	.351

For sentence reading times, the Type III Wald chi-square tests 1 showed a significant main effect of Condition, $\chi^2(2) = 18.66$, $p < .001$, but no main effect of Choice, $\chi^2(1) = 0.06$, $p = .806$, and no interaction between Condition and Choice, $\chi^2(2) = 2.10$, $p = .351$, (see Table 2). This suggests that the different focus conditions had a significant impact on the overall reading times of the sentences, but the type of choice participants made (single-object vs multi-object) did not significantly affect their reading times. In other words, while the focus type influenced how long participants took to read the sentences, it did not seem to influence their decision-making process in terms of which image they chose, nor did the choice they made influence their reading times in a way that interacted with the condition.

Table 3. *Fixed-effects tests for whole-sentence reading times, compared to exhaustive condition and the single-object image (A) choice as reference levels.*

Effect	Estimate	SE	df	p
Intercept	8.07184	0.04172	92.77	< .001
Condition (unmodified)	0.05883	0.01399	1207.28	< .001
Condition (contrastive)	0.04833	0.01991	1218.66	.015
Choice B	-0.01194	0.04849	1219.95	.806
Condition (unmodified) × Choice B	-0.09711	0.06995	1214.88	.165
Condition (contrastive) × Choice B	-0.02726	0.05422	1223.36	.615

Relative to the exhaustive condition as the reference level, whole-sentence reading times were significantly longer in both the unmodified condition ($p < .001$) and the contrastive condition ($p = .015$). This indicates that participants read exhaustive sentences faster than both unmodified PVF sentences and contrastive *amongst others*-sentences. By contrast, there was no significant effect of choice type ($p = .806$), and neither the unmodified × choice nor the contrastive × choice interaction reached significance ($p = .165$ and $p = .615$, respectively), indicating that the effect of Condition was not modulated by whether participants selected the single-object or multi-object image (see Table 3).

Post-hoc pairwise comparisons revealed that the significant effect of Condition was driven by significantly faster reading times in the exhaustive condition compared to the unmodified condition (p<.001 A–A) when participants chose the expected single-object image, but not when they chose the multi-object image (p=.994 B–B), see Table 4. There was no significant difference in whole sentence reading times between the exhaustive

Table 4. *Tukey-adjusted pairwise comparisons for whole-**sentence** reading times within choice type (A–A and B–B pairs).*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Unmodified A	-0.05883	0.0140	1212	-4.196	.0004
Exhaustive A – Contrastive A	-0.04833	0.0200	1224	-2.422	.1494
Unmodified A – Contrastive A	0.01050	0.0199	1223	0.529	.9950
Exhaustive B – Unmodified B	0.03828	0.0685	1220	0.559	.9935
Exhaustive B – Contrastive B	-0.02106	0.0492	1225	-0.428	.9982
Unmodified B – Contrastive B	-0.05935	0.0534	1221	-1.112	.8764

Table 5. *Tukey-adjusted pairwise comparisons for whole-**sentence** reading times across choice type (A–B pairs).*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Exhaustive B	0.01194	0.0486	1225	0.246	.9999
Unmodified A – Unmodified B	0.10905	0.0530	1223	2.056	.3115
Contrastive A – Contrastive B	0.03920	0.0229	1233	1.712	.5237

and contrastive conditions regardless of the choice type (p=.149 A–A, p=.998 B–B), nor between the unmodified and contrastive conditions (p=.995 A–A; p=.876 B–B). There was also no significance accross choice types within any of the conditions (see Table 5).

Taken together, this pattern of results suggests that the exhaustive focus marker *csak* 'only' facilitated processing relative to the unmodified PVF construction, but this advantage was limited to trials in which participants made the expected single-object choice. This indicates that the processing benefit of exhaustive marking emerged primarily when participants' interpretations aligned with the intended exhaustive reading, and did not extend to trials in which they selected the multi-object image. At the same time, the contrastive condition with *többek között* 'amongst others' did not differ significantly from either the exhaustive or the unmodified condition, regardless of choice type, suggesting that the contrastive modifier did not reliably affect overall reading times. Overall, any processing advantage associated with exhaustive marking appears to have been constrained rather than general. See Figure 8 for a visual representation of the whole-sentence reading times by condition, split by choice type.

3.2.2. Regions Reading Times

Region-level reading times were analyzed at three critical regions: r5 (containing the modifying phrases *csak* 'only'), a filler word, or *között*, the second half of 'amongst others', r6 (the focused object), and r7 (the sentence-final verb).

Region 5. In r5, there was a significant main effect of Condition (p < .001) (see Table 6). Using the exhaustive condition as the reference level, this effect was driven by significantly faster reading times in the exhaustive condition than in the unmodified condition (p < .001), showing that *csak* 'only' was processed more quickly than the corresponding region

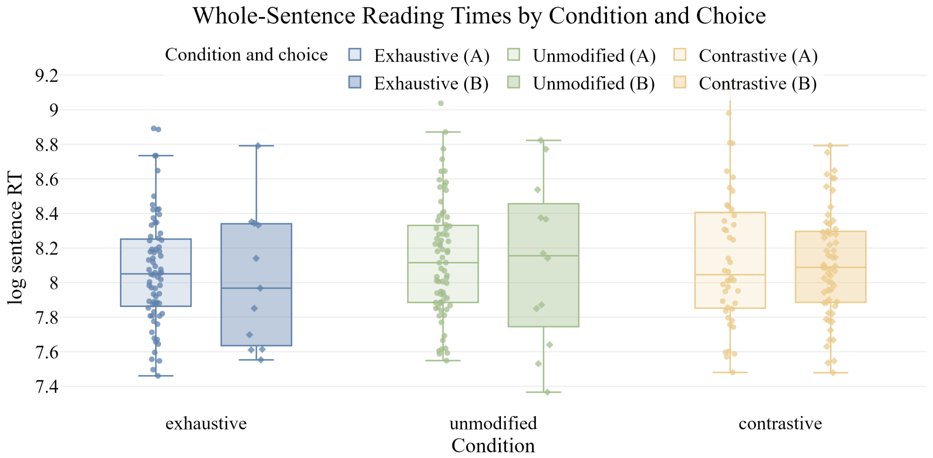

Figure 8. *Whole-sentence reading times by condition and choice type*

Table 6. *Type III Wald chi-square tests for* ***region 5*** *reading times.*

Effect	χ^2	df	*p*
Condition	18.66	2	$< .001$
Choice	0.25	1	.615
Condition × Choice	1.72	2	.423

Table 7. *Fixed effects for region 5 reading times.*

Effect	Estimate	SE	df	*p*
Intercept	6.02190	0.03691	95.57	$< .001$
Condition (unmodified)	0.06953	0.01613	1207.45	$< .001$
Condition (contrastive)	0.02961	0.02291	1226.55	.196
Chosen type B	0.02806	0.05578	1228.78	.615
Condition (unmodified) × Chosen type B	-0.10403	0.08053	1220.48	.197
Condition (contrastive) × Chosen type B	-0.05950	0.06233	1234.24	.340

in unmodified sentences. By contrast, the difference between the exhaustive and contrastive conditions was not significant ($p = .196$), indicating that r5 was not read significantly faster with *csak* ‘only’than with the contrastive modifier *többek között* ‘amongst others’. There was no significant main effect of Choice ($p = .615$), and neither interaction term reached significance (unmodified × Choice: $p = .197$; contrastive × Choice: $p = .340$), suggesting that choice type did not reliably modulate reading times in this region (see Table 7).

The Tukey-adjusted pairwise comparisons for region 5 showed that the only significant within-choice difference was between Exhaustive A and Unmodified A, with faster reading times in the exhaustive condition when participants selected image A ($p = .0003$). Across

Table 8. *Tukey-adjusted pairwise comparisons for* ***region 5*** *reading times within choice type (A–A and B–B pairs).*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Unmodified A	-0.06953	0.0162	1213	-4.303	.0003
Exhaustive A – Contrastive A	-0.02961	0.0230	1232	-1.289	.7910
Unmodified A – Contrastive A	0.03992	0.0229	1231	1.746	.5013
Exhaustive B – Unmodified B	0.03450	0.0788	1225	0.438	.9980
Exhaustive B – Contrastive B	0.02989	0.0567	1234	0.528	.9951
Unmodified B – Contrastive B	-0.00461	0.0614	1228	-0.075	1.0000

Table 9. *Tukey-adjusted pairwise comparisons for* ***region 5*** *reading times across choice type (A–B pairs).*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Exhaustive B	-0.02806	0.0559	1234	-0.502	.9961
Unmodified A – Unmodified B	0.07598	0.0611	1230	1.244	.8148
Contrastive A – Contrastive B	0.03145	0.0263	1246	1.195	.8392

choice type, the only significant contrast was between Unmodified A and Contrastive B ($p = .0021$); all other comparisons were non-significant. Overall, this suggests that the region 5 effect was driven mainly by the contrast between exhaustive and unmodified sentences in A-choice trials. This indicates that the presence of the exhaustive focus marker *csak* ‘only’ facilitated processing of the critical region compared to the unmodified PVF construction, but this advantage was specific to trials in which participants made the expected single-object choice. The processing time of the latter half of the contrastive modifier *többek között* ‘amongst others’ did not significantly differ from either the exhaustive or unmodified conditions, suggesting that it did not reliably influence reading times in this region.

Table 10. *Type III Wald chi-square tests for* ***region 6*** *reading times.*

Effect	χ^2	df	*p*
Condition	10.52	2	.005
Choice	4.52	1	.034
Condition × Choice	2.37	2	.306

Region 6. For r6, which is the focused object, there was a significant effect of Condition ($p = .005$) and Choice ($p = .034$), but no significant interaction (see Table 10). Using the exhaustive condition and Choice A as the reference levels, the effect of Condition was driven by a significant difference between the exhaustive and unmodified conditions ($p = .001$), whereas the exhaustive and contrastive conditions did not differ significantly ($p = .538$). The effect of Choice reflects longer reading times for Choice B than for Choice

Table 11. *Fixed effects for region 6 reading times.*

Effect	Estimate	SE	df	p
Intercept	6.17005	0.04615	94.31	< .001
Condition (unmodified)	0.06203	0.01942	1207.41	.001
Condition (contrastive)	0.01699	0.02760	1225.05	.538
Chosen type B	-0.14285	0.06721	1226.73	.034
Condition (unmodified) × Chosen type B	-0.01650	0.09702	1218.93	.865
Condition (contrastive) × Chosen type B	0.08434	0.07512	1231.93	.262

A across conditions (p = .034). Neither interaction term was significant (unmodified × Choice: p = .865; contrastive × Choice: p = .262), indicating that the effects of Condition and Choice were independent (see Table 11).

This suggests that the reading times for the focused object were influenced by both the focus condition and the type of choice participants made, but these effects did not interact with each other. In other words, while both the focus condition and the choice type had an impact on how long participants took to read the focused object, the way these two factors influenced reading times did not depend on each other. The significant effect of Condition indicates that the different focus conditions led to different reading times for the focused object, while the significant effect of Choice suggests that whether participants ultimately chose the single-object or multi-object image also influenced their reading times for that region.

Table 12. *Tukey-adjusted pairwise comparisons for* ***region 6*** *reading times within choice type (A–A and B–B pairs).*

Contrast	Estimate	SE	df	t	p
Exhaustive A – Unmodified A	-0.0620	0.0195	1212	-3.187	.0184
Exhaustive A – Contrastive A	-0.0170	0.0277	1230	-0.614	.9900
Unmodified A – Contrastive A	0.0450	0.0275	1229	1.635	.5752
Exhaustive B – Unmodified B	-0.0455	0.0950	1224	-0.479	.9969
Exhaustive B – Contrastive B	-0.1013	0.0683	1232	-1.485	.6742
Unmodified B – Contrastive B	-0.0558	0.0740	1227	-0.754	.9749

Table 13. *Tukey-adjusted pairwise comparisons for* ***region 6*** *reading times across choice type (A–B pairs).*

Contrast	Estimate	SE	df	t	p
Exhaustive A – Exhaustive B	0.1429	0.0674	1232	2.120	.2775
Unmodified A – Unmodified B	0.1594	0.0736	1228	2.167	.2543
Contrastive A – Contrastive B	0.0585	0.0317	1244	1.846	.4365

Pairwise comparisons showed that the significant effect of Condition in region 6 was driven by faster reading times in the exhaustive condition than in the unmodified condition ($p = .018$), whereas the exhaustive and contrastive conditions did not differ significantly ($p = .99$), possibly reflecting a residual effect of the scope marker *only*. The Tukey-adjusted comparisons further showed that, within choice type, the only significant contrast was between Exhaustive A and Unmodified A ($p = .0184$). Across choice type, significant differences emerged between Unmodified A and Exhaustive B ($p = .0278$) and between Unmodified A and Contrastive B ($p = .0001$), while all other comparisons were non-significant. Taken together, these results suggest that the effect in region 6 was driven mainly by slower reading times in the unmodified condition, especially in trials involving A choices.

Table 14. *Type III Wald chi-square tests for region 7 reading times.*

Effect	χ^2	df	p
Condition	25.48	2	< .001
Choice	0.56	1	.453
Condition × Choice	2.32	2	.314

Table 15. *Fixed effects for region 7 reading times.*

Effect	Estimate	SE	df	p
Intercept	6.21526	0.05432	93.39	< .001
Condition (unmodified)	0.11708	0.02445	1207.52	< .001
Condition (contrastive)	0.11131	0.03471	1227.70	.001
Choice B	0.06344	0.08455	1229.31	.453
Condition (unmodified) × Choice B	-0.16501	0.12209	1220.37	.177
Condition (contrastive) × Choice B	-0.12943	0.09448	1235.24	.171

Region 7. Finishing the sentence with the unstressed verb in r7, there was a significant effect of Condition ($p < .001$), but no significant effect of Choice ($p = .453$) nor a significant interaction (see Table 14). Using the exhaustive condition and Choice A as the reference, this effect of Condition was driven by significant differences between both the exhaustive and unmodified conditions ($p < .001$) and between the exhaustive and contrastive conditions ($p = .001$). On the other hand, Choice B did not differ significantly from Choice A ($p = .453$), and neither interaction term was significant (unmodified × Choice: $p = .177$; contrastive × Choice: $p = .171$), indicating that the effect of Condition was independent of participants' choice.

Post-hoc pairwise comparisons showed that the significant effect of Condition in region 7 was driven by faster reading times in the exhaustive condition than in both the unmodified condition ($p < .001$) and the contrastive condition ($p = .0177$), while the unmodified and contrastive conditions did not differ significantly. The Tukey-adjusted comparisons further indicated that these differences were restricted to A-choice trials: within

Table 16. *Tukey-adjusted pairwise comparisons for* **region 7** *reading times within choice type (A–A and B–B pairs).*

Contrast	Estimate	SE	df	t	p
Exhaustive A – Unmodified A	-0.11708	0.0245	1213	-4.779	< .0001
Exhaustive A – Contrastive A	-0.11131	0.0348	1233	-3.198	.0177
Unmodified A – Contrastive A	0.00577	0.0346	1232	0.167	1.0000
Exhaustive B – Unmodified B	0.04793	0.1200	1225	0.401	.9987
Exhaustive B – Contrastive B	0.01812	0.0859	1235	0.211	.9999
Unmodified B – Contrastive B	-0.02980	0.0931	1229	-0.320	.9996

Table 17. *Tukey-adjusted pairwise comparisons for* **region 7** *reading times across choice type (A–B pairs).*

Contrast	Estimate	SE	df	t	p
Exhaustive A – Exhaustive B	-0.06344	0.0848	1234	-0.748	.9757
Exhaustive A – Unmodified B	-0.01551	0.0924	1230	-0.168	1.0000
Exhaustive A – Contrastive B	-0.04531	0.0285	1223	-1.591	.6047
Unmodified A – Exhaustive B	0.05364	0.0844	1233	0.636	.9883
Unmodified A – Unmodified B	0.10157	0.0925	1230	1.098	.8825
Unmodified A – Contrastive B	0.07177	0.0285	1223	2.519	.1192
Contrastive A – Exhaustive B	0.04787	0.0870	1232	0.550	.9940
Contrastive A – Unmodified B	0.09580	0.0957	1232	1.001	.9176
Contrastive A – Contrastive B	0.06599	0.0399	1248	1.656	.5616

choice type, only Exhaustive A vs. Unmodified A and Exhaustive A vs. Contrastive A were significant, whereas all B–B and A–B comparisons were non-significant. Taken together, these results suggest that the effect in the final region was driven primarily by faster reading times in the exhaustive condition when participants selected image A, consistent with the idea that the final region reflects the cumulative effect of the preceding sentence on comprehension.

3.2.3. Decision Reaction Times

Table 18. *Type III Wald chi-square tests for choice reaction times.*

Effect	χ^2	df	p
Condition	5.24	2	.073
Choice	7.66	1	.006
Condition × Choice	4.30	2	.117

Table 19. *Fixed effects for choice reading times.*

Effect	Estimate	SE	df	*p*
Intercept	7.04055	0.06940	48.87	< .001
Condition (unmodified)	0.05097	0.03034	1207.63	.093
Condition (contrastive)	0.08833	0.04301	1235.44	.040
Choice B	0.29001	0.10478	1235.50	.006
Condition (unmodified) × Choice B	-0.07690	0.15146	1223.09	.612
Condition (contrastive) × Choice B	-0.22380	0.11701	1243.58	.056

When it came to decisions, the Type III Wald chi-square tests revealed a significant main effect of Choice (p = .006), but no significant main effect of Condition (p = .073) and no significant Condition × Choice interaction (p = .117) (see Table 18). Unlike the sentence reading-time measures, decision times were thus primarily affected by the type of image chosen rather than by the focus condition.

Using the exhaustive condition and Choice A as the reference levels, the model showed that decision times were significantly longer in the contrastive condition than in the exhaustive condition (p = .040), whereas the unmodified condition did not differ significantly from the exhaustive condition (p = .093). Choice B responses were significantly slower than Choice A responses (p = .006). The marginal contrastive × Choice interaction (p = .056), together with the non-significant unmodified × Choice interaction (p = .612), suggests that the A–B difference may be somewhat reduced in the contrastive condition.

Table 20. *Tukey-adjusted pairwise comparisons for* ***choice reaction times*** *within choice type (A–A and B–B pairs).*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Unmodified A	-0.0510	0.0304	1213	-1.676	.5478
Exhaustive A – Contrastive A	-0.0883	0.0431	1241	-2.049	.3154
Unmodified A – Contrastive A	-0.0374	0.0429	1239	-0.870	.9535
Exhaustive B – Unmodified B	0.0259	0.1480	1228	0.175	1.0000
Exhaustive B – Contrastive B	0.1355	0.1060	1241	1.273	.7997
Unmodified B – Contrastive B	0.1095	0.1150	1234	0.949	.9335

Table 21. *Tukey-adjusted pairwise comparisons for* ***choice reaction times*** *across choice type (A–B pairs).*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Exhaustive B	-0.2900	0.1050	1241	-2.761	.0646
Unmodified A – Unmodified B	-0.2131	0.1150	1236	-1.858	.4290
Contrastive A – Contrastive B	-0.0662	0.0493	1261	-1.343	.7607

Table 22. *Type-III Wald χ^2 tests from mixed models.*

Metric	Effect	χ^2	df	p
total_dwell	condition	12.531	2	.0019 **
	choice	11.845	1	.0006 ***
	condition × choice	12.292	2	.0021 **
total_fixation	condition	12.326	2	.0021 **
	choice	3.346	1	.0674 .
	condition × choice	6.764	2	.0340 *

The Tukey-adjusted pairwise comparisons refine this pattern. Within Choice A, no between-condition comparison reached significance after adjustment. Within Choice B, no pairwise differences between conditions were significant (all $ps \geq .800$). Within conditions, none of the A–B contrasts reached significance after Tukey correction: exhaustive (p = .065), unmodified (p = .429), and contrastive (p = .761). Taken together, decision times for A and B choices did not differ significantly within any condition, suggesting that for participants both readings were broadly available regardless of the focus type. It is also important that in the unmodified condition, decision times did not differ significantly between A and B choices, suggesting that for participants both readings might are equally possible.

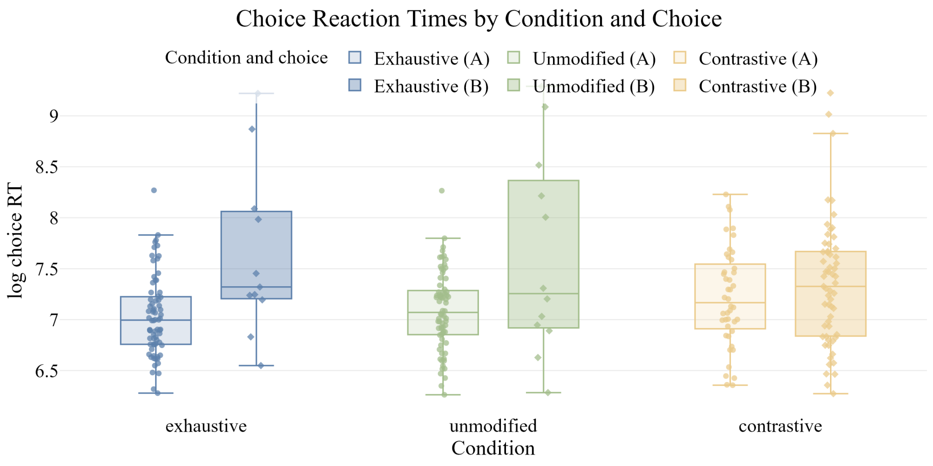

Figure 9. *Choice reaction times by condition, split by choice*

3.3. Eye-tracking results

To further examine how the focus structures were associated with exhaustive and contrastive readings, and how the conditions and choices influenced cognitive processing, we

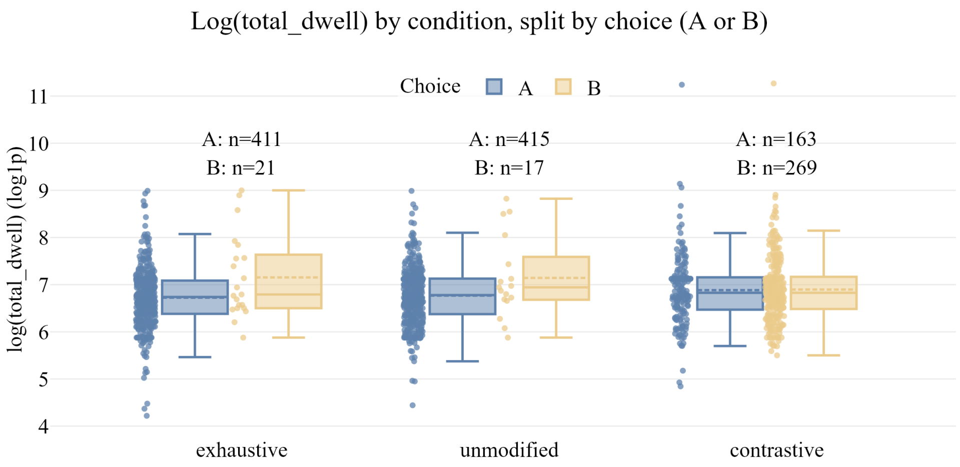

Figure 10. *Total dwell time (log-transformed) by condition and choice type*

fitted linear mixed-effects models to two eye-tracking metrics: *total dwell time*, and *total fixation count* with fixed effects of Condition (exclusive–*only*, unmodified, and contrastive–*among others*), Choice (single- vs multi-object type images), and their interaction, with random intercepts for Participant and Item. Dependent variables were log-transformed using `log1p` here as well. Type-III Wald χ^2 tests and Tukey-adjusted pairwise comparisons from `emmeans` were used for inference.

3.3.1. Dwell times

Table 23. *Type III Wald chi-square tests for total dwell time.*

Effect	χ^2	df	p
Condition	12.53	2	.002
Chosen type	11.85	1	< .001
Condition × Chosen type	12.29	2	.002

For *total dwell time* – the total duration of all gaze episodes on the AOIs during the choice – we found significant main effects of Condition: $\chi^2(2) = 12.531$, $p = 0.00190$ and Choice: $\chi^2(1) = 11.845$, $p = 0.000578$, as well as a significant interaction of Condition×Choice: $\chi^2(2) = 12.292$, $p = 0.00214$ (Table 23).

Using the exhaustive condition and choice A as the reference levels, total dwell time was significantly higher in the contrastive condition ($\beta = 0.18755$, $p < .001$) and for choice B ($\beta = 0.44427$, $p < .001$), whereas the unmodified condition did not differ significantly from the exhaustive baseline ($\beta = 0.04735$, $p = .208$). There was no significant interaction between the unmodified condition and choice B ($\beta = -0.20522$, $p = .273$), but there was a significant negative interaction between the contrastive condition and choice B ($\beta =$

Table 24. *Fixed effects for total dwell time.*

Effect	Estimate	SE	df	p
Intercept	6.72603	0.06414	54.54	< .001
Condition (unmodified)	0.04735	0.03756	1208.21	.208
Condition (contrastive)	0.18755	0.05298	1252.59	< .001
Choice B	0.44427	0.12909	1253.64	< .001
Condition (unmodified) × Choice B	-0.20522	0.18700	1234.61	.273
Condition (contrastive) × Choice B	-0.47795	0.14392	1264.64	< .001

-0.47795, $SE = 0.14392$, $df = 1264.64$, $p < .001$), indicating that the positive effect of choice B was substantially reduced in the contrastive condition. Overall, both condition and choice type affected total dwell time, and their effects were not simply additive (Table 24).

Table 25. *Tukey-adjusted pairwise comparisons for total dwell times within choice type (A–A and B–B pairs).*

Contrast	Estimate	SE	df	t	p
Exhaustive A – Unmodified A	-0.0474	0.0376	1213	-1.258	.8077
Exhaustive A – Contrastive A	-0.1876	0.0531	1258	-3.530	.0057
Unmodified A – Contrastive A	-0.1402	0.0529	1256	-2.650	.0864
Exhaustive B – Unmodified B	0.1579	0.1830	1239	0.862	.9553
Exhaustive B – Contrastive B	0.2904	0.1310	1259	2.215	.2315
Unmodified B – Contrastive B	0.1325	0.1420	1249	0.931	.9386

Table 26. *Tukey-adjusted pairwise comparisons for total dwell times across choice type (A–B pairs).*

Contrast	Estimate	SE	df	t	p
Exhaustive A – Exhaustive B	-0.4443	0.1290	1259	-3.432	.0081
Unmodified A – Unmodified B	-0.2390	0.1410	1251	-1.689	.5390
Contrastive A – Contrastive B	0.0337	0.0605	1283	0.557	.9937

Post-hoc (Tukey-adjusted) comparisons indicated that in the exhaustive condition, participants have spent significantly less time looking at the AOIs when they chose the A-type, single-object image (A–B differed: $\hat{\beta} = -0.4443$, SE $= 0.1290$, $t(1259) = -3.432$, $p = 0.0006$) signifying that the expected choice was not only easy but also a fast one, which makes sense for this control-like condition (see Figure **??**). Follow-up pairwise comparisons also showed that participants who ultimately commited to the A-type, single-object image have spent significantly more time looking at the AOIs in the contrastive condition than in the exhaustive and unmodified focus conditions (see Table **??**). This suggests that the contrastive condition elicited more cognitive effort and uncertainty when participants

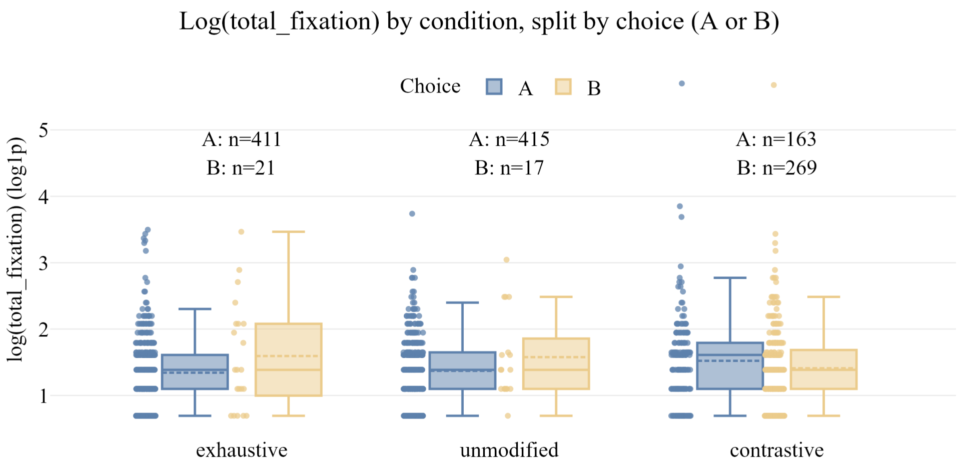

Figure 11. *Total fixation count (log-transformed) by condition and choice type*

chose the single-object images, meaning that they have pondered longer on the decision, presumably finding it a bit more difficult. In contrast, for those who ended up choosing the B-type, multi-object image, there was no significant difference between the conditions, except a marginal trend for less gaze times in the contrastive condition when pitted against the exhaustive sentences.

3.3.2. Fixation count

Table 27. *Type III Wald chi-square tests for total fixation times.*

Effect	χ^2	df	*p*
Condition	12.33	2	.002
Chosen type	3.35	1	.067
Condition × Chosen type	6.76	2	.034

Table 28. *Fixed effects for total fixation times.*

Effect	Estimate	SE	df	*p*
Intercept	1.34642	0.05092	69.60	< .001
Condition (unmodified)	0.02654	0.03208	1208.20	.408
Condition (contrastive)	0.15751	0.04525	1252.55	< .001
Choice B	0.20164	0.11023	1254.25	.068
Condition (unmodified) × Choice B	-0.07622	0.15967	1235.25	.633
Condition (contrastive) × Choice B	-0.28643	0.12290	1265.34	.020

For *total fixation count*, the trends were similar but much weaker. The Type III Wald chi-square tests showed a significant main effect of Condition on fixation count, $\chi^2(2) = 12.33$, $p = .002$, and a significant Condition × Choice interaction, $\chi^2(2) = 6.76$, $p = .034$, but no significant main effect of Choice, $\chi^2(1) = 3.35$, $p = .067$ (see Table 27). Thus, fixation counts varied across conditions, and the effect of condition depended on which image type participants chose.

The fixed-effects model, using the exhaustive condition and Choice A as reference levels, showed that fixation counts were significantly higher in the contrastive condition ($\beta = 0.15751$, $SE = 0.04525$, $df = 1252.55$, $p < .001$), whereas the unmodified condition did not differ from the exhaustive baseline ($\beta = 0.02654$, $SE = 0.03208$, $df = 1208.20$, $p = .408$; see Table 28). Choice B showed only a marginal effect ($\beta = 0.20164$, $SE = 0.11023$, $df = 1254.25$, $p = .068$). There was no significant interaction between the unmodified condition and Choice B ($\beta = -0.07622$, $SE = 0.15967$, $df = 1235.25$, $p = .633$), but there was a significant negative interaction between the contrastive condition and Choice B ($\beta = -0.28643$, $SE = 0.12290$, $df = 1265.34$, $p = .020$), indicating that the increase in fixation count in the contrastive condition was reduced when participants selected Choice B.

Table 29. *Tukey-adjusted pairwise comparisons for total fixation times within choice type (A–A and B–B pairs).*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Unmodified A	-0.02654	0.0321	1213	-0.826	.9628
Exhaustive A – Contrastive A	-0.15751	0.0454	1258	-3.471	.0071
Unmodified A – Contrastive A	-0.13097	0.0452	1256	-2.898	.0441
Exhaustive B – Unmodified B	0.04968	0.1560	1239	0.318	.9996
Exhaustive B – Contrastive B	0.12892	0.1120	1260	1.151	.8596
Unmodified B – Contrastive B	0.07924	0.1220	1249	0.652	.9869

Table 30. *Tukey-adjusted pairwise comparisons for total fixation times across choice type (A–B pairs).*

Contrast	Estimate	SE	df	*t*	*p*
Exhaustive A – Exhaustive B	-0.20164	0.1110	1259	-1.824	.4507
Unmodified A – Unmodified B	-0.12542	0.1210	1252	-1.038	.9051
Contrastive A – Contrastive B	0.08479	0.0517	1284	1.640	.5718

The post-hoc comparisons within choice type showed that for Choice A, fixation counts were significantly lower in the exhaustive condition than in the contrastive condition ($\beta = -0.15751$, $p = .0071$), and also lower in the unmodified condition than in the contrastive condition ($\beta = -0.13097$, $t = -2.898$, $p = .0441$). By contrast, exhaustive and unmodified Choice A trials did not differ significantly ($p = .9628$). For Choice B, none of the condition contrasts reached significance (all $p \geq .8596$), indicating that the condition effect was driven primarily by trials in which participants selected Choice A.

The across-choice comparisons showed no significant difference between Choice A and Choice B within any single condition: exhaustive A vs. exhaustive B was not significant ($\beta = -0.20164$, $p = .4507$), nor were unmodified A vs. unmodified B ($p = .9051$) and contrastive A vs. contrastive B ($p = .5718$). Taken together, these post-hoc results suggest that the interaction observed in reflects differences across conditions for Choice A, rather than robust within-condition differences between Choice A and Choice B.

4. Discussion

4.1. Overview

The central question of this study was whether the Hungarian pre-verbal focus (PVF) construction inherently encodes exhaustivity, or whether exhaustive and contrastive readings are both available, arising from different pragmatic inferences. Our results, drawn from behavioural choice data, self-paced reading times, and webcam-based eye tracking, converge on a consistent picture: the PVF construction carries a strong but cancellable exhaustive bias. This bias is not semantically obligatory, but it is robust – strong enough to persist even in the presence of an explicit contrastive licensing phrase. The following sections discuss the implications of each converging line of evidence, and then situate these findings within the broader theoretical debate and prior experimental literature.

4.2. The exhaustive bias of the unmodified PVF construction

The most striking finding from the behavioural data is perhaps also the most expected one, yet it has not been demonstrated this directly before: unmodified PVF sentences pattern almost identically with explicitly exhaustive *csak* 'only'-sentences in terms of choice behaviour. Participants chose the single-object, exhaustive interpretation in 95.1% of exhaustive trials and in 96.1% of unmodified PVF trials (see Table 1), with no significant difference between the two. This near-ceiling alignment is striking and helps explain the intuition that motivated the original claim by É.Kiss (1998) and later by Szabolcsi (1981) and Kenesei (2006) that the PVF position is inherently exhaustive: for everyday, out-of-context sentences, the exhaustive reading is overwhelmingly dominant, and it is understandable why this was originally taken to be a semantic, rather than pragmatic, property.

However, the similarity in choice rates between the exhaustive and unmodified conditions does not mean these two conditions were processed identically. The reading time data reveals a qualitative difference: exhaustive sentences were read significantly faster than unmodified PVF sentences at the sentence level ($p < .001$), and at each of the critical regions – the modifier/focus region (r5, $p = .0003$), the focused object (r6, $p = .018$), and the sentence-final verb (r7, $p < .0001$). This processing advantage was consistently observed in trials where participants ultimately selected the expected single-object image (Choice A), but not in multi-object choice trials, suggesting that the facilitation was tied to the alignment between the explicit exhaustive reading and the chosen interpretation, rather than to the exhaustive marker *csak* per se.

This pattern is consistent with an account in which explicitly marked exhaustivity (*csak* 'only') provides an early, unambiguous cue that guides comprehension, allowing readers to resolve the sentence meaning faster and with less processing cost. In contrast, an unmodified PVF sentence leaves the degree of exclusivity initially underspecified, requiring the reader to derive the exhaustive interpretation via pragmatic inference – and this inferential step comes at a measurable processing cost. This interpretation aligns directly with Balogh (2013)'s proposal that exhaustivity in the PVF is not due to a covert exhaustivity operator, but is derived as a pragmatic implicature, similarly to how scalar inferences arise in the broader experimental pragmatics literature (cf. Káldi and Babarczy 2018). The analogy with scalar implicature computation is theoretically informative: just as the inference from 'some' to 'not all' is computed later and more effortfully than a lexically specified quantifier (cf. Káldi et al. 2021), the exhaustive reading of a bare PVF sentence appears to require an additional inferential step compared to the sentence with *csak*.

4.3. Contrastive readings: available but costly

The contrastive condition reveals the complementary side of this picture. When the PVF was modified with *többek között* 'amongst others' – a phrase with inherently inclusive, non-exhaustive meaning – participants switched to preferring the multi-object, contrastive interpretation in 62.3% of trials, a dramatic shift from the near-floor level of B-choices in the other two conditions (4.86% and 3.94% respectively; see Table 1 and Figure 6). This clearly demonstrates that a contrastive, non-exhaustive reading of the PVF construction is not only theoretically possible but is genuinely available to speakers and can be readily elicited. This is an important empirical finding, corroborating the theoretical positions of Wedgwood et al. (2006), Onea & Beaver (2009), Balogh (2013), and Káldi and Babarczy (2016), who all challenged the view that PVF obligatorily encodes exhaustivity.

Notably, however, the contrastive reading did not dominate the contrastive condition either: 37.7% of choices in this condition were still the single-object, exhaustive image. This is a significant proportion. Even when the sentence explicitly licences a non-exhaustive reading with a contrastive modifier, the association between the PVF construction and exhaustive interpretation is strong enough to persist at non-trivial levels. The fact that exhaustivity is 'cancellable' in the formal, pragmatic sense – as shown by the grammaticality of sentences such as (4) – does not mean that cancellation is cognitively effortless. These results suggest that the exhaustive bias associated with PVF is deeply entrenched in speakers' representations of this construction, such that even an explicit licensing phrase does not fully override it.

This finding is further corroborated by the reaction time and eye-tracking data for the contrastive condition. Decision reaction times were significantly longer in the contrastive condition than in the exhaustive condition ($p = .040$), and – crucially – participants who ultimately chose the single-object image (A) in the contrastive condition spent significantly more time looking at the images before deciding than participants in the exhaustive condition ($\hat{\beta} = 0.1876$, $p = .0057$, Table 25). Choosing an exhaustive image after reading a sentence that grammatically licences a non-exhaustive reading is thus a cognitively costly act – one that requires overcoming the pragmatic push of *többek között*. Similarly, fixation counts for Choice A were significantly higher in the contrastive than in either the exhaustive ($p = .0071$) or unmodified ($p = .0441$) conditions. Together, these results are consistent with the idea that committing to the exhaustive interpretation in a contrastive context requires additional deliberation, reflecting the competition between the PVF's default exhaustive bias and the non-exhaustive semantics of the modifier.

4.4. The unmodified PVF as semantically underspecified

Perhaps the most theoretically significant result from the eye-tracking analysis is the absence of a significant difference in total dwell time between Choice A and Choice B in the *unmodified* condition ($\hat{\beta} = -0.2390$, $p = .539$; see Table 26). This stands in stark contrast to the exhaustive condition, where participants dwelled significantly less on the images when selecting the expected exhaustive (A) choice ($\hat{\beta} = -0.4443$, $p = .0081$), reflecting the ease and clarity of the exhaustive decision. In the unmodified condition, however, participants showed no significant asymmetry in how long they looked before choosing A or B. This null result is informative: it suggests that, during the actual moment of decision, both the exhaustive and contrastive interpretations were equally plausible to participants when the PVF was used without any modifying marker.

This pattern of results is consistent with a model of semantic underspecification (cf. Wedgwood et al. 2006), in which the unmodified PVF does not lexically or structurally fix the degree of exhaustivity, leaving this to be resolved by contextual and pragmatic factors. The strong behavioural preference for single-object images (96.1% Choice A) reflects a default pragmatic bias – a prior expectation that PVF sentences are exhaustive in the absence of other information – but the eye-tracking data suggests that this default is not so deeply entrenched as to make the contrastive reading cognitively inaccessible. In other words, the majority of participants converge on the exhaustive reading, but those who entertained the contrastive reading did so without appreciably greater cognitive effort. The PVF, in isolation, is thus best characterised not as a semantic exhaustivity operator, but as a pragmatically enriched construction with a strong, context-sensitive exhaustive default.

This result reconciles what initially appear to be contradictory observations in the literature. The received view (É.Kiss 1998; Horvath 2008; Kenesei 2006; Szabolcsi 1981) is motivated by the near-universal intuition that PVF sentences entail exhaustivity – which our data confirms. But the conclusion that this entailment is semantic rather than pragmatic was based on intuitions and theoretical argument, not on direct evidence about the availability of non-exhaustive readings. Our results, along with the earlier experimental work by Onea & Beaver (2009), Gerőcs et al. (2014), and Káldi and Babarczy (2018), provide such evidence, and it consistently points toward a pragmatic analysis.

4.5. Relation to prior experimental work

Our findings converge with and extend a growing body of experimental literature on Hungarian focus exhaustivity. Onea & Beaver (2009) were among the first to provide experimental evidence that the Hungarian PVF is not semantically exhaustive, showing that exhaustivity effects are stronger in the presence of verbal particles (and thus cannot be attributed to focus position alone), and proposing a pragmatic analysis grounded in question-answering. Our results are fully consistent with their core claim and add direct evidence from a richer set of dependent measures – not only behavioural choices, but also reading times and eye movements – that allow us to characterise the temporal dynamics and cognitive cost of exhaustive and contrastive interpretations.

Gerőcs et al. (2014) provided further experimental evidence using an offline task paradigm, showing that exhaustive interpretation of the PVF is not obligatory and can be contextually modulated. Our study extends this with an online, time-sensitive paradigm that tracks both reading and gaze, thereby providing a window into the comprehension process rather than just its endpoint. The directional pattern in our data aligns with theirs: unmodified PVF is exhaustive by default, but not obligatorily so.

Most directly comparable to the current study is Káldi and Babarczy (2018), who used a visual-world eye-tracking paradigm to investigate how context (restrictive vs. non-restrictive) modulates the exhaustive inference associated with Hungarian PVF. Their central finding was that the rate of exhaustive interpretation was context-dependent, supporting a pragmatic/implicature account of PVF exhaustivity. Our study complements this by removing context manipulation and instead comparing conditions that explicitly force or cancel exhaustivity against an unmodified baseline, making it possible to characterise the strength and resilience of the exhaustive default. The null A–B dwell difference in our unmodified condition, together with the strong choice preference for A, mirrors the picture Káldi and Babarczy describe: PVF creates a strong exhaustive tendency, but this tendency is sensitive to linguistic input and is not a hard semantic entailment. Káldi (2021)'s doctoral work situating exhaustivity as a scalar type implicature provides a natural theoretical home for these empirical patterns: like other scalar inferences, the exhaustive reading is the pragmatically expected default, but can be computed more slowly than lexically specified exclusivity and is cancellable by appropriate context or explicit licensing.

Our findings also invite a cross-linguistic perspective. Hsu (2019) investigated the association between focus constructions and exhaustivity in Chinese, comparing wh-focus, cleft-focus, and *only*-focus using a trichotomous-response design. Their results showed that cleft-focus and only-focus were clearly distinguished from wh-focus in terms of exhaustivity associations, with both triggering robust exhaustive readings, while still differing subtly from each other in conscious decision-making tasks. The structurally different mechanism of Hungarian PVF – syntactic fronting and eradicating prosodic stress, as opposed to a cleft construction – nonetheless leads to a broadly similar functional outcome: a construction that strongly associates with exhaustivity without necessarily encoding it as a semantic entailment. This parallel across typologically distinct languages is consistent with the view that exhaustive readings of grammaticalized focus constructions arise through a common pragmatic mechanism, rather than through a language-specific semantic operator, a point also raised by Krifka (2008) in his account of information structure subtypes.

4.6. Theoretical implications

Taken together, our results have clear implications for the theoretical debate on the semantics and pragmatics of Hungarian PVF. The strong semantic exhaustivity view – that an immediately pre-verbal focused constituent is semantically exhaustive, i.e. it is true that no other entity satisfies the predicate (É.Kiss 1998, 2007; Horvath 2008; Kenesei 2006; Szabolcsi 1981) – predicts that (i) non-exhaustive readings should be unavailable or highly marked, (ii) the unmodified PVF should be processed identically to the *only*-condition, and (iii) co-occurrence with *többek között* 'amongst others' should be degraded or induce a strong semantic conflict. None of these predictions is borne out. Non-exhaustive readings were chosen in over a third of contrastive trials, the unmodified condition was processed more slowly than the explicitly marked exhaustive condition, and the contrastive condition was grammatically acceptable and produced meaningful, interpretable results.

In contrast, the pragmatic exhaustivity view – most fully developed for Hungarian by Balogh (2013) in the tradition of Rooth (1992)'s alternative semantics and Vallduví & Vilkuna (1998)'s notion of *kontrast* – predicts that (i) the exhaustive reading is the default inference but can be cancelled, (ii) the lexical exhaustivity marker provides a shorter, more direct path to the exhaustive interpretation, and (iii) cancellation of the exhaustive reading (via *többek között*) is possible but comes at a processing cost, given the competing default. All three of these predictions are supported by our data. Zimmermann (2008)'s view that contrastivity is a pragmatic, discourse-structuring phenomenon, rather than a semantic property, is also consistent: the PVF does not select between exhaustive and contrastive readings at the semantic level; instead, this is a discourse-level resolution that depends on context, prior beliefs, and the presence of additional cues.

In É.Kiss's (1998) framework, both exhaustive and contrastive readings fall under *identificational focus*, defined as a focus that identifies a referent from a set of alternatives, described as [+exhaustive] and optionally [+contrastive]. Our results challenge the claim that such focus is inherently [+exhaustive] as a semantic entailment, while being compatible with the weaker claim that exhaustivity is the prototypical inferential outcome of identificational focus. Indeed, the very near-ceiling exhaustive choices for the unmodified PVF (96.1%) show that the pragmatic bias towards exhaustivity is extremely strong – strong enough to look, at first, like a semantic entailment – but the eye-tracking evidence of equal decision effort for A and B choices in this condition reveals that the construction leaves the exhaustivity question genuinely open.

It is also worth connecting these results to Gécseg's (2020) observation that focused topics in Hungarian show discourse-configurational prominence without necessarily encoding exhaustivity, and to Genzel et al. (2015)'s finding that prosodic expression in Hungarian focus constructions varies as a function of the type of focus (narrow, broad, contrastive). These latter results suggest that even at the phonological level, the PVF construction is not a monolithic exhaustivity trigger; the same syntactic position can host focus with distinct prosodic, semantic, and pragmatic profiles. Our results add the comprehension dimension to this picture, showing that the same surface form – the unmodified PVF – is genuinely compatible with more than one interpretive outcome.

4.7. *Limitations and future directions*

Several limitations of the current study deserve acknowledgement. First, our eye-tracking data was collected via webcam using the WebGazer.js library integrated into PCIbex, which, while enabling large-scale online data collection, has substantially lower spatial and temporal precision than laboratory-based eye-tracking systems. Gaze was classified into two broad areas of interest (left and right canvases), collapsing the rich spatio-temporal information available in high-resolution systems. Our episode-based cleaning procedure addressed some artefacts of this method, but absolute magnitudes of dwell time and fixation count should be interpreted cautiously, and the null effects we report (particularly in the unmodified condition) should be understood in the context of this reduced precision. Nonetheless, the consistency between the dwell time and fixation count results, and their alignment with the reading time and behavioural patterns, strengthens confidence in the key contrasts.

Second, the forced-choice, sentence–picture verification paradigm we used requires participants to commit to one of two interpretations and does not capture gradient or intermediate representations. The 37.7% exhaustive choices in the contrastive condition, for example, could reflect either genuine exhaustive interpretations or a residual structural bias that the paradigm forces into a categorical outcome. Future work using magnitude estimation, acceptability rating, or more fine-grained response measures could help disentangle these possibilities.

Third, the contrastive condition employed a single licensing phrase, *többek között* 'amongst others'. It remains an open question whether other contrastive devices – additive particles, correction contexts, or naturally occurring discourse embedding – would produce the same degree of exhaustivity suppression or a different pattern. The robustness of the exhaustive default across different contrastive contexts is an empirically open question that deserves further investigation.

Fourth, as an online study with participant-owned devices, there was considerable variability in the technical quality of the eye-tracking data. A substantial proportion of participants could not pass the calibration threshold and were thus excluded before the experiment began. This raises a self-selection concern: participants who successfully calibrated may not be fully representative of the broader population of Hungarian speakers, potentially skewing the sample toward those with higher-resolution webcams, better lighting, or greater attentiveness to technical instructions.

Future research could address these limitations in several ways. A laboratory-based replication with high-resolution remote eye-tracking would allow more precise characterisation of the gaze dynamics, including measures such as first-fixation latency, proportion of looks over time, and fixation duration distributions. The use of context-embedded stimuli – placing the target PVF sentence in a prior discourse that either restricts or expands the relevant set of alternatives – would allow a more direct comparison with Káldi and Babarczy (2018)'s context-dependent exhaustivity paradigm. Production experiments could complement the comprehension data by examining which conditions prompt speakers to spontaneously add exhaustive or contrastive modifiers to an otherwise bare PVF. Finally, a systematic cross-linguistic investigation using the same paradigm for other languages with grammaticalized focus constructions (cf. Cruschina 2021; Hsu 2019) would allow researchers to test whether the strong-but-cancellable exhaustive bias is a universal property of syntactically marked identificational focus, or a Hungarian-specific effect shaped by the language's particular discourse-configurational grammar (É. Kiss 1995).

Besides this, it would be interesting to investigate whether specific features of Hungarian have any effect on the interpretation of focus constructions, such the definiteness–indefiniteness of the objects and the use of the definite conjugation, the countability–uncountability of the objects, the presence of verbal prefixes, the aspect of the verbs, and the tense of the verbs (past vs present).

5. Conclusion

This study investigated the interpretation and processing of Hungarian pre-verbal focus (PVF) constructions using a sentence–picture verification task with self-paced reading and webcam-based eye tracking. Our central empirical question was whether the PVF construction inherently encodes exhaustivity, or whether both exhaustive and contrastive, non-exclusive readings are genuinely available to Hungarian speakers. The results provide a clear and consistent answer: the PVF construction does not independently encode exhaustivity, but it carries a strong and robust default bias towards exhaustive interpretation.

Behaviourally, unmodified PVF sentences were interpreted exhaustively in over 96% of trials – virtually identically to sentences with the explicit exhaustive marker *csak* 'only' – demonstrating that the exhaustive reading is the overwhelming default in the absence of contextual information. This confirms the intuition behind the classical semantic accounts (É.Kiss 1998; Kenesei 2006; Szabolcsi 1981). However, it does not vindicate the stronger semantic claim: when the PVF was modified with *többek között* 'amongst others', a phrase that licences a non-exhaustive, contrastive reading, participants shifted to preferring the multi-object image in 62.3% of trials, clearly showing that the exhaustive reading is cancellable and that a contrastive interpretation is genuinely available. The exhaustivity bias persisted even here – roughly a third of contrastive-condition choices were still exhaustive – but it was substantially weakened.

The processing data add an important dimension that the behavioural choices alone cannot reveal. Reading times were fastest for the explicitly exhaustive condition at every critical region, reflecting the processing efficiency of a lexically specified, unambiguous exhaustivity cue. The unmodified PVF was read more slowly, consistent with the view that exhaustivity must be inferred pragmatically rather than read off directly from the semantics of the construction. Most tellingly, eye-tracking data showed that participants deliberated equally long before choosing an exhaustive or contrastive image for unmodified PVF sentences – a null result that is informative: it suggests that both readings remained equally plausible at the point of decision. In contrast, exhaustive-condition choices were made faster and with less hesitation, and contrastive-condition choices for the unexpected exhaustive image required significantly more deliberation and more looks at the images.

Taken together, these findings support the view that PVF exhaustivity arises not from a covert semantic operator or structural feature, but from a pragmatic inference – a strong but contextually sensitive default that can be manipulated, cancelled, and bears a measurable processing cost when overridden. This positions the current study alongside and in support of the growing experimental literature challenging the semantic exhaustivity view (Gerőcs et al. 2014; Káldi and Babarczy 2018; Káldi 2021; Onea & Beaver 2009), and provides the most direct online psycholinguistic evidence to date that the Hungarian PVF construction is semantically underspecified with respect to exhaustivity.

More broadly, these results demonstrate that the long-standing theoretical controversy about whether PVF exhaustivity is semantic or pragmatic can and should be resolved through psycholinguistic experimentation. The parallel processing cost profiles for pragmatic inference – slower reading, greater deliberation, more effortful decisions – converge with what experimental pragmatics has established for scalar implicatures more generally (cf. Káldi et al. 2021), supporting a unified account in which exhaustive readings of PVF sentences are a species of pragmatic enrichment, not truth-conditional content. The construction is, in a meaningful sense, ambiguous – or at least underspecified – and Hungarian speakers navigate this underspecification fluidly, defaulting to exhaustivity but remaining sensitive to linguistic cues that license contrastive alternatives.

6. NOTES

6.1. Todo list

- Check Discussion section again.

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Data availability statement. Stimuli, full results, replication data and code can be found at https://osf.io/j9abg/overview?view_only=484a4f21a5cb4e51b4b826f970e096a0.

Appendix

Figure 12. *Complete visual stimulus set with A) single- and B) multi-entity image pairs*

Table 31.: Full stimulus list with sentence regions

no	item	condition	r1	r2	r3	r4	r5	r6	r7
1	1	exhaustive	*Ami*	*a gyümölcsöket*	*illeti,*	*Anna*	*csak*	*egy banánt*	*evett.*
			which (relative)	the fruits-ACC	regards	Anna	only	a banana-ACC	ate-3SG
			‘Regarding fruits, Anna ate only a banana.’						
2	1	unmodified	*Ami*	*a gyümölcsöket*	*illeti,*	*Anna*	*tegnap*	*egy banánt*	*evett.*
			which (relative)	the fruits-ACC	regards	Anna	yesterday	a banana-ACC	ate-3SG
			‘Regarding fruits, Anna ate a banana yesterday.’						
3	1	contrastive	*Tegnap*	*reggel*	*Anna*	*többek*	*között*	*egy banánt*	*evett.*
			yesterday	morning	Anna	others	among	a banana-ACC	ate-3SG
			‘Yesterday morning Anna ate, among other things, a banana.’						
4	2	exhaustive	*Ami*	*az írószereket*	*illeti,*	*a kisfiú*	*csak*	*egy ceruzát*	*kapott.*
			which (relative)	the stationeries-ACC	regards	the boy	only	a pencil-ACC	received-3SG
			‘Regarding stationery, the boy received only a pencil.’						
5	2	unmodified	*Ami*	*az írószereket*	*illeti,*	*a kisfiú*	*tegnap*	*egy ceruzát*	*kapott.*
			which (relative)	the stationeries-ACC	regards	the boy	yesterday	a pencil-ACC	received-3SG
			‘Regarding stationery, the boy received a pencil yesterday.’						
6	2	contrastive	*Tegnap*	*reggel*	*a kisfiú*	*többek*	*között*	*egy ceruzát*	*kapott.*
			yesterday	morning	the boy	others	among	a pencil-ACC	received-3SG
			‘Yesterday morning the boy received, among other things, a pencil.’						
7	3	exhaustive	*Ami*	*a bútorokat*	*illeti,*	*az asztalos*	*csak*	*egy könyvespolcot*	*készített.*
			which (relative)	the furniture-ACC	regards	the carpenter	only	a bookshelf-ACC	made-3SG
			‘Regarding furniture, the carpenter made only a bookshelf.’						
8	3	unmodified	*Ami*	*a bútorokat*	*illeti,*	*az asztalos*	*tegnap*	*egy könyvespolcot*	*készített.*
			which (relative)	the furniture-ACC	regards	the carpenter	yesterday	a bookshelf-ACC	made-3SG
			‘Regarding furniture, the carpenter made a bookshelf yesterday.’						
9	3	contrastive	*Tegnap*	*a műhelyben*	*az asztalos*	*többek*	*között*	*egy könyvespolcot*	*készített.*
			yesterday	in the workshop	the carpenter	others	among	a bookshelf-ACC	made-3SG
			‘Yesterday in the workshop the carpenter made, among other things, a bookshelf.’						
10	4	exhaustive	*Ami*	*a harci eszközöket*	*illeti,*	*a katonaság*	*csak*	*a helikoptert*	*tesztelte.*
			which (relative)	the military equipment-ACC	regards	the military	only	the helicopter-ACC	tested-3SG
			‘Regarding the equipment, the military only tested the helicopter.’						
11	4	unmodified	*Ami*	*a harci eszközöket*	*illeti,*	*a katonaság*	*tavaly*	*a helikoptert*	*tesztelte.*

Continued on next page

no	item	condition	r1	r2	r3	r4	r5	r6	r7
			which (relative)	the military equipment-ACC	regards	the military	last year	the helicopter-ACC	tested-3SG
			‘Regarding the equipment, the military tested the helicopter this time last year.’						
12	4	contrastive	*Tavaly*	*ilyenkor*	*a katonaság*	*többek*	*között*	*a helikoptert*	*tesztelte.*
			last year	this time	the military	others	among	the helicopter-ACC	tested-3SG
			‘This time last year the military tested, among other things, the helicopter.’						
13	5	exhaustive	*Ami*	*a süteményeket*	*illeti,*	*én*	*csak*	*a fánkot*	*sütöttem.*
			which (relative)	the sweets-ACC	regards	I	only	the doughnut-ACC	baked-1SG
			‘Regarding the sweets, I only baked the doughnut.’						
14	5	unmodified	*Ami*	*a süteményeket*	*illeti,*	*én*	*múltkor*	*a fánkot*	*sütöttem.*
			which (relative)	the sweets-ACC	regards	I	last time	the doughnut-ACC	baked-1SG
			‘Regarding the sweets, I baked the doughnut last time.’						
15	5	contrastive	*Tegnap*	*este*	*én*	*többek*	*között*	*a fánkot*	*sütöttem.*
			yesterday	evening	I	others	among	the doughnut-ACC	baked-1SG
			‘Yesterday evening I baked, among other things, a doughnut.’						
16	6	exhaustive	*Ami*	*a virágokat*	*illeti,*	*Éva*	*csak*	*a tulipánt*	*szereti.*
			which (relative)	the flowers-ACC	regards	Eve	only	the tulip-ACC	likes-3SG
			‘Regarding flowers, Eve only likes tulips.’						
17	6	unmodified	*Ami*	*a virágokat*	*illeti,*	*Éva*	*mostanában*	*a tulipánt*	*szereti.*
			which (relative)	the flowers-ACC	regards	Eve	nowadays	the tulip-ACC	likes-3SG
			‘Regarding flowers, Eve likes tulips nowadays.’						
18	6	contrastive	*Hallottam*	*hogy*	*Éva*	*többek*	*között*	*a tulipánt*	*szereti.*
			heard-1SG	that	Eve	others	among	the tulip-ACC	likes-3SG
			‘I heard that Eve likes, among other things, tulips.’						
19	7	exhaustive	*Ami*	*a bevásárlást*	*illeti,*	*mi*	*csak*	*ásványvizet*	*vettünk.*
			which (relative)	the shopping-ACC	regards	we	only	mineral water-ACC	bought-1PL
			‘Regarding shopping, we only bought mineral water.’						
20	7	unmodified	*Ami*	*a bevásárlást*	*illeti,*	*mi*	*tegnap*	*ásványvizet*	*vettünk.*
			which (relative)	the groceries-ACC	regards	we	yesterday	mineral water-ACC	bought-1PL
			‘Regarding shopping, we bought mineral water yesterday.’						
21	7	contrastive	*Tegnap*	*este*	*mi*	*többek*	*között*	*ásványvizet*	*vettünk.*
			yesterday	evening	we	others	among	mineral water-ACC	bought-1PL
			‘Yesterday evening we bought, among other things, mineral water.’						
22	8	exhaustive	*Ami*	*az innivalót*	*illeti,*	*Eszter*	*csak*	*kávét*	*ivott.*
			which (relative)	the drink-ACC	regards	Eszter	only	coffee-ACC	drank-3SG

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no	item	condition	r1	r2	r3	r4	r5	r6	r7
			‘Regarding drinks, Eszter only drank coffee.’						
23	8	unmodified	*Ami*	*az innivalót*	*illeti,*	*Eszter*	*ma*	*kávét*	*ivott.*
			which (relative)	the drink-ACC	regards	Eszter	today	coffee-ACC	drank-3SG
			‘Regarding drinks, Eszter drank coffee today.’						
24	8	contrastive	*Ma*	*reggel*	*Eszter*	*többek*	*között*	*kávét*	*ivott.*
			today	morning	Eszter	others	among	coffee-ACC	drank-3SG
			‘This morning, Eszter drank, among other things, coffee.’						
25	9	exhaustive	*Ami*	*a nyelveket*	*illeti,*	*a diákok*	*csak*	*németet*	*tanultak.*
			which (relative)	the languages-ACC	regards	the students	only	German-ACC	learned-3PL
			‘Regarding languages, the students only learned German.’						
26	9	unmodified	*Ami*	*a nyelveket*	*illeti,*	*a diákok*	*régen*	*németet*	*tanultak.*
			which (relative)	the languages-ACC	regards	the students	long ago	German-ACC	learned-3PL
			‘Regarding languages, students learned German in the past.’						
27	9	contrastive	*Régen*	*az iskolában*	*a diákok*	*többek*	*között*	*németet*	*tanultak.*
			long ago; in the past	at school	the students	others	among	German-ACC	learned-3PL
			‘In the past, at school, students learned, among other things, German.’						
28	10	exhaustive	*Ami*	*az új fajokat*	*illeti,*	*Marci*	*csak*	*egy lepkét*	*fedezett fel.*
			which (relative)	the new species-ACC	regards	Marci	only	a butterfly-ACC	discovered-3SG
			‘Regarding new species, Marci discovered only a butterfly.’						
29	10	unmodified	*Ami*	*az új fajokat*	*illeti,*	*Marci*	*tavaly*	*egy lepkét*	*fedezett fel.*
			which (relative)	the new species-ACC	regards	Marci	last year	a butterfly-ACC	discovered-3SG
			‘Regarding new species, last year Marci discovered a butterfly.’						
30	10	contrastive	*Tavaly*	*nyáron*	*Marci*	*többek*	*között*	*egy lepkét*	*fedezett fel.*
			last year	summer	Marci	others	among	a butterfly-ACC	discovered-3SG
			‘Last summer Marci discovered, among other things, a butterfly.’						
31	11	exhaustive	*Ami*	*a járműveket*	*illeti,*	*a turisták*	*csak*	*egy robogót*	*béreltek ki.*
			which (relative)	the vehicles-ACC	regards	the tourists	only	a scooter-ACC	hired-3PL
			‘Regarding vehicles, the tourists hired only a scooter.’						
32	11	unmodified	*Ami*	*a járműveket*	*illeti,*	*a turisták*	*a hétvégén*	*egy robogót*	*béreltek ki.*
			which (relative)	the vehicles-ACC	regards	the tourists	on the weekend	a scooter-ACC	hired-3PL
			‘Regarding vehicles, on the weekend the tourists hired a scooter.’						
33	11	contrastive	*Múlt*	*hétvégén*	*a turisták*	*többek*	*között*	*egy robogót*	*béreltek ki.*
			last	weekend	the tourists	others	among	a scooter-ACC	hired-3PL
			‘Last weekend, the tourists hired, among other things, a scooter.’						

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no	item	condition	r1	r2	r3	r4	r5	r6	r7
34	12	exhaustive	*Ami* which (relative)	*a kertet* the garden-ACC	*illeti,* regards	*a férfi* the man	*csak* only	*egy fenyőfát* a pine tree-ACC	*ültetett el.* planted-3SG
			‘Regarding the garden, the man planted only a pine tree.’						
35	12	unmodified	*Ami* which (relative)	*a kertet* the garden-ACC	*illeti,* regards	*a férfi* the man	*tíz éve* ten years ago	*egy fenyőfát* a pine tree-ACC	*ültetett el.* planted-3SG
			‘Regarding the garden, the man planted a pine tree ten years ago.’						
36	12	contrastive	*Tíz évvel ez előtt* ten years ago	*a kertben* in the garden	*a férfi* the man	*többek* others	*között* among	*egy fenyőfát* a pine tree-ACC	*ültetett el.* planted-3SG
			‘Ten years ago the man planted, among other things, a pine tree.’						
37	13	exhaustive	*Ami* which (relative)	*az állatokat* the animals-ACC	*illeti,* regards	*a gyerekek* the children	*csak* only	*az elefántot* the elephant-ACC	*nézték meg.* looked at-3PL
			‘Regarding animals, the children looked at only the elephant.’						
38	13	unmodified	*Ami* which (relative)	*az állatokat* the animals-ACC	*illeti,* regards	*a gyerekek* the children	*múltkor* last time	*az elefántot* the elephant-ACC	*nézték meg.* looked at-3PL
			‘Regarding animals, the children looked at the elephant last time.’						
39	13	contrastive	*Múltkor* last time	*az állatkertben* in the zoo	*a gyerekek* the children	*többek* others	*között* among	*az elefántot* the elephant-ACC	*nézték meg.* looked at-3PL
			‘Last time in the zoo the children looked at, among other things, the elephant.’						
40	14	exhaustive	*Ami* which (relative)	*a konyhát* the kitchen-ACC	*illeti,* regards	*a szerelő* the technician	*csak* only	*a hűtőt* the fridge-ACC	*üzemelte be.* installed-3SG
			‘Regarding the kitchen, the technician installed only the fridge.’						
41	14	unmodified	*Ami* which (relative)	*a konyhát* the kitchen-ACC	*illeti,* regards	*a szerelő* the technician	*múltkor* last time	*a hűtőt* the fridge-ACC	*üzemelte be.* installed-3SG
			‘Regarding the kitchen, the technician installed the fridge last time.’						
42	14	contrastive	*Múltkor* last time	*a konyhában* in the kitchen	*a szerelő* the technician	*többek* others	*között* among	*a hűtőt* the fridge-ACC	*üzemelte be.* installed-3SG
			‘Last time in the kitchen the technician installed, among other things, the fridge.’						
43	15	exhaustive	*Ami* which (relative)	*a kincseket* the treasures-ACC	*illeti,* regards	*a kalóz* the pirate	*csak* only	*az aranyat* the gold-ACC	*találta meg.* found-3SG
			‘Regarding the treasures, the pirate only found the gold.’						
44	15	unmodified	*Ami* which (relative)	*a kincseket* the treasures-ACC	*illeti,* regards	*a kalóz* the pirate	*éjjel* at night	*az aranyat* the gold-ACC	*találta meg.* found-3SG
			‘Regarding the treasures, the pirate found the gold at night.’						
45	15	contrastive	*Éjjel* at night	*a szigeten* on the island	*a kalóz* the pirate	*többek* others	*között* among	*az aranyat* the gold-ACC	*találta meg.* found-3SG

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no	item	condition	r1	r2	r3	r4	r5	r6	r7
			‘At night on the island the pirate found, among other things, the gold.’						
46	16	exhaustive	*Ami*	*az ebédet*	*illeti,*	*a nagyi*	*csak*	*levest*	*tálalt fel.*
			which (relative)	the lunch-ACC	regards	the grandma	only	soup-ACC	served-3SG
			‘Regarding lunch, grandma served only soup.’						
47	16	unmodified	*Ami*	*az ebédet*	*illeti,*	*a nagyi*	*ma*	*levest*	*tálalt fel.*
			which (relative)	the lunch-ACC	regards	the grandma	today	soup-ACC	served-3SG
			‘Regarding lunch, grandma served soup today.’						
48	16	contrastive	*Ma*	*ebédre*	*a nagyi*	*többek*	*között*	*levest*	*tálalt fel.*
			today	for lunch	the grandma	others	among	soup-ACC	served-3SG
			‘For today’’s lunch, grandma served, among other things, soup.’						
49	17	exhaustive	*Ami*	*a ruhákat*	*illeti,*	*ez a szabó*	*csak*	*ingeket*	*foltozott be.*
			which (relative)	the clothes-ACC	regards	this tailor	only	shirts-ACC	patched-3SG
			‘Regarding clothes, this tailor patched only shirts.’						
50	17	unmodified	*Ami*	*a ruhákat*	*illeti,*	*ez a szabó*	*általában*	*ingeket*	*foltozott be.*
			which (relative)	the clothes-ACC	regards	this tailor	usually	shirts-ACC	patched-3SG
			‘Regarding clothes, this tailor usually patched shirts.’						
51	17	contrastive	*Általában*	*a műhelyben*	*ez a szabó*	*többek*	*között*	*ingeket*	*foltozott be.*
			usually	in the workshop	this tailor	others	among	shirts-ACC	patched-3SG
			‘Usually in the workshop this tailor patched, among other things, shirts.’						
52	18	exhaustive	*Ami*	*a táncot*	*illeti,*	*Péter*	*csak*	*Marit*	*kérte fel.*
			which (relative)	the dance-ACC	regards	Peter	only	Mari-ACC	asked-3SG
			‘Regarding the dance, Peter asked only Mary.’						
53	18	unmodified	*Ami*	*a táncot*	*illeti,*	*Péter*	*tegnap este*	*Marit*	*kérte fel.*
			which (relative)	the dance-ACC	regards	Peter	last night	Mari-ACC	asked-3SG
			‘Regarding the dance, Peter asked Mary last night.’						
54	18	contrastive	*Tegnap este*	*a bálon*	*Péter*	*többek*	*között*	*Marit*	*kérte fel.*
			last night	at the ball	Peter	others	among	Mari-ACC	asked-3SG
			‘Last night at the ball, Peter asked, among others, Mary [for a dance].’						

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